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PHOTONIC COUPLERS FOR ELECTRONIC/PHOTONIC PACKAGES AND METHODS OF FORMING THE SAME

Abstract

An embodiment photonic interconnect die may include a substrate, a dielectric waveguide, including a core portion and a cladding portion, formed on the substrate, a first photonic coupler formed at a first end of the dielectric waveguide, and a second photonic coupler formed at a second end of the dielectric waveguide. The dielectric waveguide may include a planar geometry within the cladding portion such that a surface of the dielectric waveguide is parallel to a first surface of the photonic interconnect die. The first photonic coupler and the second photonic coupler may each be configured to couple photonic signals into and out of the photonic interconnect die such that a photonic signal pathway connects the first photonic coupler, the dielectric waveguide, and the second photonic coupler. The photonic interconnect die may further couple photonic signals into and out of a first dielectric window and a second dielectric window.

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Background/Summary

BACKGROUND

[0001] Many computing applications use optical (i.e., photonic) signals to provide secure high-speed data transmission. Various emerging technologies are also being developed that may provide functionality to perform computing operations directly on optical/photonic signals. Silicon photonics is a promising technology area that uses semiconductor device processing techniques to provide systems including integrated electronic and photonic components. Such components may be used for the generation, routing, modulation, processing, and detection of light. Together, these functions form a photonic analog to electronic integrated circuits (EIC) and, as such, may constitute photonic integrated circuits (PIC).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. **1** is a schematic illustration of various components that may be used in a photonic computing system.

[0004] FIG. **2** is a schematic top view of an example photonic integrated circuit.

[0005] FIG. **3**A is vertical cross-sectional view of a photonic integrated circuit formed on a silicon-on-insulator (SOI) substrate, according to various embodiments.

[0006] FIG. **3**B is vertical cross-sectional view of a portion of the photonic integrated circuit of FIG. **3**A showing photonic components, according to various embodiments.

[0007] FIG. **3**C is vertical cross-sectional view of a further portion of the photonic integrated circuit of FIG. **3**A showing a photonic coupler, according to various embodiments.

[0008] FIG. **3**D is vertical cross-sectional view of a further portion of the photonic integrated circuit of FIG. **3**A showing a further photonic coupler, according to various embodiments.

[0009] FIG. **4** is a vertical cross-sectional view of an electronic/photonic package including an optical engine and a separate electronic component, according to various embodiments.

[0010] FIG. **5**A is a top view of a further electronic/photonic package including a photonic interconnect die, according to various embodiments.

[0011] FIG. **5**B is a vertical cross-sectional view of the electronic/photonic package of FIG. **5**A, according to various embodiments.

[0012] FIG. **6**A is a top view of a photonic interconnect die that photonically connects a first photonic component and a second photonic component, according to various embodiments.

[0013] FIG. **6**B is a top view of a photonic interconnect die that photonically connects a first photonic component, a second photonic component, a third photonic component, and a fourth photonic component, according to various embodiments.

[0014] FIG. **7** is a vertical cross-sectional view of a further electronic/photonic package including a photonic interconnect die, according to various embodiments.

[0015] FIG. **8**A is a top view of a further electronic/photonic package including a first photonic interconnect die, a second photonic interconnect die, and a third photonic interconnect die, according to various embodiments.

[0016] FIG. **8**B is a vertical cross-sectional view of the electronic/photonic package of FIG. **8**A, according to various embodiments.

[0017] FIG. **8**C is a three-dimensional perspective view showing details of a fiber array unit, according to various embodiments.
[0018] FIG. **9**A is a top view of a further electronic/photonic package including a plurality of photonic interconnect dies, according to various embodiments.
[0019] FIG. **9**B is a vertical cross-sectional view of a portion of the electronic/photonic package of FIG. **9**A, according to various embodiments.
[0020] FIG. **10**A is a top view of a further electronic/photonic package including a plurality of photonic interconnect dies, according to various embodiments.
[0021] FIG. **10**B is a vertical cross-sectional view of a portion of the electronic/photonic package of FIG. **10**A, according to various embodiments.
[0022] FIG. **11** is a top view of a further electronic/photonic package including a first component package and a second component package, according to various embodiments.
[0023] FIG. **12** is a vertical cross-sectional view of a further electronic/photonic package including a photonic interconnect die having edge couplers, according to various embodiments.
[0024] FIG. **13** is a vertical cross-sectional view of a further electronic/photonic package including a photonic interconnect die that couples photonic signals between two hybrid interposers, according to various embodiments.
[0025] FIG. **14**A is a vertical cross-sectional view of an intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0026] FIG. **14**B is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0027] FIG. **14**C is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0028] FIG. **14**D is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0029] FIG. **14**E is a top view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0030] FIG. **15**A is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0031] FIG. **15**B is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0032] FIG. **15**C is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0033] FIG. **16**A is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0034] FIG. **16**B is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0035] FIG. **16**C is a vertical cross-sectional view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0036] FIG. **16**D is a vertical cross-sectional view of a photonic interconnect die, according to various embodiments.
[0037] FIG. **16**E is a vertical cross-sectional view of a further photonic interconnect die, according to various embodiments.
[0038] FIG. **16**F is a vertical cross-sectional view of a further photonic interconnect die, according to various embodiments.
[0039] FIG. **16**G is a vertical cross-sectional view of a further photonic interconnect die, according to various embodiments.
[0040] FIG. **16**H is a vertical cross-sectional view of a further photonic interconnect die, according to various embodiments.
[0041] FIG. **17**A is a top view of a photonic interconnect die including edge couplers, according to

various embodiments.
[0042] FIG. **17**B is a vertical cross-sectional view of the photonic interconnect die of FIG. **17**A, according to various embodiments.
[0043] FIG. **18**A is a top view of an intermediate structure that may be used to form a photonic interconnect die including grating couplers, according to various embodiments.
[0044] FIG. **18**B is a vertical cross-sectional view of the intermediate structure of FIG. **18**A, according to various embodiments.
[0045] FIG. **18**C is a vertical cross-sectional view of a photonic interconnect die including grating couplers, according to various embodiments.
[0046] FIG. **19**A is a top view of an intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0047] FIG. **19**B is a top view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0048] FIG. **19**C is a top view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0049] FIG. **19**D is a top view of a further intermediate structure that may be used to form a photonic interconnect die, according to various embodiments.
[0050] FIG. **20** is a flowchart illustration operations of a method of forming a photonic interconnect die, according to various embodiments.

DETAILED DESCRIPTION

[0051] The following disclosure provides many different embodiments, or examples, for implementing unique features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
[0052] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Unless explicitly stated otherwise, each element having the same reference numeral is presumed to have the same material composition and to have a thickness within a same thickness range.
[0053] Various embodiments disclosed herein provide photonic interconnect dies that allow photonic signals to be propagated between a first photonic component and a second photonic component in an electronic/photonic package. The incorporation of optical/photonic signaling functionality into an electronic/photonic package may provide reduced signal delay and ohmic loss. In this regard, in certain embodiments, to avoid long electrical signal pathways and associated delay and ohmic loss, it may be advantageous to convey signals in the form of photonic signals along a portion of the signal pathway. This may be done by converting electrical signals into photonic signals at a first point along the signal pathway, propagating the photonic signals for a certain distance, and then converting the photonic signals back into electrical signals at a second point along the signal pathway. In still further embodiments, the functionality to convert electrical

signals into photonic signals and vice versa may be advantageous in photonic (quantum or classical) computing operations. The use of photonic interconnect dies allows various electronic and photonic components to be fabricated separately as stand-alone dies. Such dies may then be assembled into an electronic/photonic package and photonic components may be coupled by photonic interconnect dies.

[0054] An embodiment photonic interconnect die may include a substrate, a dielectric waveguide, including a core portion and a cladding portion, formed on the substrate, a first photonic coupler formed at a first end of the dielectric waveguide, and a second photonic coupler formed at a second end of the dielectric waveguide. The dielectric waveguide may include a planar geometry within the cladding portion such that a surface of the dielectric waveguide is parallel to a first surface of the photonic interconnect die. The first photonic coupler and the second photonic coupler may each be configured to couple photonic signals into and out of the photonic interconnect die such that a photonic signal pathway connects the first photonic coupler, the dielectric waveguide, and the second photonic coupler. The photonic interconnect die may further couple photonic signals into and out of a first dielectric window and a second dielectric window.

[0055] According to a further embodiment, an electronic/photonic package is provided. The electronic/photonic package may include a first photonic component including first photonic signal pathways, a second photonic component including second photonic signal pathways, and a photonic interconnect die including a plurality of dielectric waveguides. The photonic interconnect die may be coupled to the first photonic component and the second photonic component such that the first photonic signal pathways are photonically coupled to the second photonic signal pathways by the plurality of dielectric waveguides.

[0056] An embodiment method of forming a photonic interconnect die may include forming a waveguide cladding portion on a substrate; forming a waveguide core portion within the waveguide cladding portion; forming a first photonic coupler at a first end of the waveguide core portion; and forming a second photonic coupler at a second end of the waveguide core portion.

[0057] FIG. **1** is an illustration of various components that may be used in a photonic computing system. System components may include a generation device also referred to as a photonic source **102** such as a laser or light-emitting diode (LED), a routing device that may include a plurality of dielectric waveguides **104** configured to route photonic signals, and a detector that includes one or more photonic detectors **106** configured to detect photonic signals and to convert received photonic signals into output electrical signals. Additional components may include a modulation device that includes one or more photonic modulators **108** and photonic processing components **110**.

[0058] The one or more photonic modulators **108** may take an input electronic signal and may modulate an input photonic signal to impose an amplitude and/or phase modulation on the input photonic signal in response to an input electronic signal. In this way, the one or more photonic modulators **108** may be used to convert data provided in the form of an electronic signal into data encoded as a photonic signal. Similarly, the one or more photonic detectors **106** may convert processed photonic signals back into output electrical signals. The photonic processing components **110** may be configured to perform (classical or quantum) logic operations on the modulated photonic signal. The various photonic components (**102** to **110**) may be integrated within a single die to thereby form a photonic integrated circuit (PIC), as described in greater detail with reference to FIGS. **2** to **3**D, below.

[0059] FIG. **2** is a schematic top view of an example PIC **200**, according to various embodiments. The PIC **200** may include photonic sources **102**, first photonic couplers **112***a*, dielectric waveguides **104**, beam splitters **114**, photonic modulators **108**, a photonic multiplexer **116**, second photonic couplers **112***b*, and one or more photonic detectors **106**. Other components (not shown) may include active or passive photonic amplifiers, photonic switches, (quantum or classical) logic gates, etc. The PIC **200** of FIG. **2** may be configured as a photonic transceiver that may generate a plurality of photonic signals along a transmitter path **202***a* and may receive phonic signals along a receiver path **202**_b_. In other embodiments (not shown) the PIC may be configured as a transmitter (i.e., omitting the receiver path **202**_b_) or as a receiver (i.e., omitting the transmitting path **202**_a_). Various other PIC devices (not shown) may provide various other types of functionality such as allowing logic operations to be performed directly on photonic signals.

[0060] In the example embodiment PIC **200** of FIG. **2**, the transmitter path **202**_a_ may be configured to receive input electrical signals from various input electrical connections (e.g., certain ones of a plurality of electrical circuit pads **118**) and may generate output photonic signals that may be provided to one or more output channels, such as output optical fibers **204**_a_. The receiver path **202**_b_ may receive various input photonic signals from one or more input channels, such as input optical fibers **204**_b_ and may convert the received input photonic signals to corresponding output electrical signals that may be provided to various output electrical connections (e.g., other ones of the plurality of electrical circuit pads **118**).

[0061] According to various embodiments, the transmitter path **202**_a_ may be configured as follows. Each photonic source **102** may include a laser or LED that may generate an unmodulated signal, such as a continuous wave (CW) light/radiation beam. The unmodulated signal generated by each photonic source **102** may then be provided to a respective first photonic coupler **112**_a_, which may then provide the unmodulated signal to a dielectric waveguide **104** and to a photonic modulator **108**. As shown, beam splitters **114** may also be used to increase a number of signals provided to the photonic modulators **108**. Each photonic modulator **108** may then generate a modulated signal from a respective received unmodulated signal in response to a time-dependent input electrical signal (e.g., received from some of the electrical circuit pads **118**). The modulated signals may then be fed to the multiplexer **116** by additional dielectric waveguides **104**. The multiplexer **116** may then generate multiplexed output signal that may be fed to respective output channels, such as the output optical fibers **204**_a_.

[0062] The various photonic sources **102** may generate a plurality of unmodulated signals having a corresponding plurality of respective wavelengths. The multiplexer **116** may then combine various modulated signals having different wavelengths into a smaller number of output photonic channels that may carry the multiplexed signals. Other photonic system components (e.g., photonic processing components **110**) may then receive the multiplexed photonic signals and may use a de-multiplexer (not shown) to separate (i.e., de-multiplex) the various photonic signals for further processing operations.

[0063] As shown in FIG. **2**, the receiver path **202**_b_ may include one or more photonic detectors **106**. For simplicity of description, only a single photonic detector **106** is shown. According to various embodiments, however, the PIC **200** may include a plurality of input optical fibers **204**_b_ that may receive a plurality of corresponding input photonic signals. In some embodiments, the input optical signals may include multiplexed data including photonic signals having a plurality of wavelengths. A de-multiplexer (not shown) may then separate the various signals having respective wavelengths. Each separated signal may then be provided to a respective photonic detector **106**, which may convert the received photonic signal into a corresponding output electrical signal. The resulting output electrical signal may then be provided as output to other ones of the electrical circuit pads **118**.

[0064] Some of the components of the PIC **200** may be passive components (e.g., beam splitters **114**, dielectric waveguides **104**, multiplexer **116**, etc.) that do not generate or receive electrical signals. Other components of the PIC **200** may be active components (e.g., photonic sources **102**, photonic detectors **106**, modulators **108**, etc.) that may receive or generate electrical signals. As such, the PIC **200** may include both electrical and photonic circuits. As described above, the PIC **200** may include a plurality of electrical circuit pads **118** that may be electrically connected to the various active components of the PIC **200**. As such, some of the plurality of electrical circuit pads **118** may be configured to receive input electrical signals and other ones of the plurality of electrical circuit pads **118** may be configured to provide output electrical signals. Still further other ones of the plurality of electrical circuit pads **118** may be configured to receive electrical power that may be provided to the active components that require a power supply (e.g., the photonic sources **102**).

[0065] According to various embodiments, the electrical and photonic circuit elements of the PIC **200** may be formed using semiconductor processing techniques in various front-end-of-line (FEOL) and back-end-of-line (BEOL) operations. As such, the PIC **200** may be fabricated as a stand-alone photonic die that may be incorporated into a photonic package structure, as described in greater detail below. In this regard, various electro-optic circuit elements (e.g., the photonic sources **102** and photonic detectors **106**) may include semiconductor device components that may be fabricated at a semiconductor substrate level in a FEOL process as well as at various interconnect levels in a BEOL process. For example, in some embodiments, certain active components may include control circuits including transistor structures (e.g., CMOS circuits) formed in a FEOL process.

[0066] Various electrical interconnect structures and other transistor structures may be formed in a BEOL process. Active and passive components may be formed within or above various interconnect levels in BEOL processes. For example, electro-optical components (e.g., photonic sources **102**, photonic modulators **108**, photonic detectors **106**, etc.) may be formed in BEOL processes and may include, for example, thin-film transistors that may include, for example, oxide semiconductor materials. Passive components, such as dielectric waveguides **104**, may be formed by deposition and pattering of various dielectric structures. For example, a dielectric waveguide **104** may be formed to include a core material having a higher index of refraction than a surrounding cladding material, as described in greater detail below.

[0067] FIG. **3**A is vertical cross-sectional view of a PIC **200** formed on a silicon-on-insulator (SOI) substrate **302**, according to various embodiments. The SOI substrate **302** may include a silicon substrate layer **302***a*, an oxide layer **304** formed over the silicon substrate layer **302***a*, and a silicon layer **302***b* formed over the oxide layer **304**. As described above, semiconductor device fabrication techniques (e.g., lithographic patterning and etching) may be performed on the silicon layer **302***b* to thereby form photonic components (**104**, **106**, **108**) and photonic couplers (**112***a*, **112***b*).

[0068] As shown in FIG. **3**A, a vertically oriented optical fiber **204***v* may be coupled to the PIC **200** in a nominally vertical orientation, as described in greater detail with reference to FIG. **3**C, below. Alternatively, a horizontally oriented optical fiber **204***h* may be coupled to the PIC **200**, as described in further detail with reference to FIG. **3**D, below. The PIC **200** may also include various electrical circuit elements (not shown) that may also be formed along with the photonic components (**104**, **106**, **108**) and photonic couplers (**112***a*, **112***b*) using semiconductor device fabrication techniques. As described above with reference to FIG. **2**, the electrical circuit elements may provide electrical power to active photonic components and may allow input electrical signals to be provided to the PIC **200** and for output electrical signals to be received from the PIC **200** by other components.

[0069] FIG. **3**B is vertical cross-sectional view of a portion **300***b* of the PIC **200** of FIG. **3**A showing photonic components (**104**, **106**, **108**), according to various embodiments. FIG. **3**C is vertical cross-sectional view of a further portion **300***c* of the PIC **200** of FIG. **3**A showing a first photonic coupler **112***a*, according to various embodiments, and FIG. **3**D is vertical cross-sectional view of a further portion **300***d* of the photonic integrated circuit of FIG. **3**A showing a second photonic coupler **112***b*, according to various embodiments. As shown in FIG. **3**B, the photonic components (**104**, **106**, **108**) may include dielectric waveguides **104**, photonic detectors **106**, photonic modulators **108**, etc. Other photonic components (not shown) may include photonic sources **102**, splitters **114**, multiplexers **116**, etc.

[0070] Photonic signals may be coupled into and out from the PIC **200** by various photonic couplers (**112***a*, **112***b*). As shown in FIG. **3**C, the first photonic coupler **112***a* may be a grating coupler **306***a* that may allow coupling between the PIC **200** and a vertically oriented optical fiber **204***v*. Further, as shown in FIG. **3**D, the second photonic coupler **112***b* may be configured as an edge coupler **306***b* that may allow coupling between the PIC **200** and a horizontally oriented optical fiber **204***h*. The first photonic coupler **112***a* may be formed by etching a periodic array of shallow grooves **308** in the dielectric waveguide **104**. Every such shallow groove in the periodic array may act as a scatterer of electromagnetic radiation (i.e., light/photons). As such, the first photonic coupler **112***a* may be configured as a diffraction grating. Such a diffraction grating may be configured such that scattered contributions from the various groves interfere constructively in predetermined direction (e.g., 8-10 degrees relative to an upward of a top surface of the PIC **200**). In this way, a photonic signal **310** (i.e., an electromagnetic wave) may be coupled from the first photonic coupler **112***a* to the vertically oriented optical fiber **204***v* as shown in FIG. **3**C. Similarly photonic signals (not shown) may be coupled from the vertically oriented optical fiber **204***v* into the PIC **200** by way of the first photonic coupler **112***a*.

[0071] As shown in FIG. **3**D, the second photonic coupler **112***b* may be configured as an edge coupler **306***b*. In this regard, a photonic signal **310** (i.e., light/photons) may be coupled out of a surface of a dielectric waveguide **104** (also referred to as "butt-coupling") and into the horizontally oriented optical fiber **204***h*. According to various embodiments, however, there may be a disparity in optical mode size between the dielectric waveguide **104** and the horizontally oriented optical fiber **204***h*. In this regard, the optical mode size refers to a spatial extent of electro-magnetic fields in directions perpendicular (e.g., along the z-axis) to a longitudinal axis (e.g., along the x-axis) of the dielectric waveguide **104**. For example, a silicon dielectric waveguide **104** may have an optical mode size that is between 3-4 microns, while a mode size of an optical fiber may be between 8-10 microns.

[0072] To accommodate for the disparity in mode sizes between the dielectric waveguide **104** and the horizontally oriented optical fiber **204***h*, the second photonic coupler **112***b* may include a spot-size-converter (SSC). The SSC (not shown) may have a lateral profile that gradually changes size with distance along the propagation direction (e.g., along the x-axis in FIG. **3**D). By slowly changing (e.g., increasing) the lateral size of the SSC, light propagating within the SSC may be confined in the fundamental optical mode with a spot size being gradually expanded to match a size of a fiber core (not shown) of the horizontally oriented optical fiber **204***h*. To reduce optical insertion loss, an undercut structure **312** may be provided to prevent the expanded optical mode propagating within the SSC from overlapping with the silicon substrate layer **302***a* underneath the SSC.

[0073] As shown in FIG. **3**D, the undercut structure **312** may be a region free of silicon under the second photonic coupler **112***b* and may be formed by performing a two-step etching process. In a first operation, openings (not shown) may be etched in the SOI substrate **302**. The openings may then provide a conduit for etchant chemicals, in a wet etching process, to reach a silicon surface at an interface between the silicon substrate layer **302***a* and the oxide layer **304** in a second etching process. In this regard, the silicon may be isotropically etched through the openings. A dielectric material may then be formed in the undercut structure **312** to provide structural stability to the undercut structure **312**. The dielectric material may be chosen to have a refractive index sufficiently different from that of silicon such that the optical mode does not propagate within the undercut region **312**.

[0074] FIG. **4** is a vertical cross-sectional view of an electronic/photonic package **400** including an optical engine **402** and an electronic component **406**, according to various embodiments. The optical engine **402** may be configured to convert electrical signals into photonic signals **310** and vice versa. The electronic component **406** may be an active EIC such as a CPU die, a GPU die, an application specific integrated circuit (ASIC), a memory die, etc. Alternatively, the electronic component **406** may be a passive component such as a capacitor, an inductor, as resistor, a diode, a transformer, an integrated passive die, etc.

[0075] The optical engine **402** and the electronic component **406** may each be attached to, and electrically coupled to, an electronic interposer **408**. Each of the optical engine **402** and the electronic component **406** may further receive electrical signals from, and may transmit electrical signals to, the electronic interposer **408**. In this regard, the optical engine **402** and the electronic component **406** may communicate with one another by way of electrical connections provided by the electronic interposer **408**. The optical engine **402** and the electronic component **406** may be attached to the electronic interposer **408** with a plurality of first solder material portions **407***a*. The electronic interposer **408** may, in turn, be attached to, and electrically coupled to, a package substrate **410** with a plurality of second solder material portions **407***b*. The package substrate **410** may allow the electronic/photonic package **400** to be connected to other system components such as a printed circuit board (PCB) (not shown).

[0076] The optical engine **402** may be used to reduce signal delay and ohmic loss by providing an optical/photonic signaling functionality. Thus, in certain embodiments, to avoid long electrical signal pathways and associated delay and ohmic loss, it may be advantageous to convey signals in the form of photonic signals along a portion of the signal pathway. This may be done by converting electrical signals into photonic signals at a first point along the signal pathway, propagating the photonic signals for a certain distance, and then converting the photonic signals back into electrical signals at a second point along the signal pathway. In still further embodiments, the functionality to convert electrical signals into photonic signals and vice versa may be advantageous in photonic (quantum or classical) computing operations, as described in greater detail with reference to FIGS. **9**A and **9**B, below.

[0077] The optical engine **402** may be configured as a co-packaged electronic/photonic die. In this regard, an EIC **404** may be bonded to a PIC **200** with hybrid bonding structures **405**. As described above with reference to FIGS. **2** to **3**D, the PIC **200** may include electrical and optical/photonic circuits that may be configured to generate photonic signals **310** based on received electrical signals. The optical engine **402** may also receive photonic signals **310** and may process the received photonic signals **310** to generate electrical signals that may be provided to the electronic interposer **408** as output.

[0078] The EIC **404** may be configured to provide control signals to the PIC **200**. For example, the EIC **404** may provide a modulated electrical signal that may encode digital information. The PIC **200** may then use various electro-optic components to generate photonic signals **310** based on the modulated electrical signal received from the EIC **404**. For example, the PIC **200** may include one or more photonic modulators **108** and one or more dielectric waveguides **104**. The PIC **200** may further include one or more optical couplers **112** that may convey photonic signals **310** between the PIC **200** and an integrated optics section **412** of the optical engine **402**. The integrated optics section **412** may include various optical components such as lenses **414**, reflectors (not shown), diffraction gratings (also not shown), etc. The integrated optics section **412** may further include various coupling structures (e.g., a fiber array unit (not shown)) that may optically couple the integrated optics section **412** to one or more optical fibers. For example, as shown in FIG. **4**, the integrated optics section **412** may be mechanically and optically coupled to a nominally vertically oriented optical fiber **204***v*.

[0079] As shown in FIG. **4**, the electronic interposer **408** may include a plurality of electrical interconnects (**416***a*, **416***b*) formed within one or more dielectric layers (**418***a*, **418***b*). In various embodiments, the electronic interposer **408** may be a semiconductor interposer, a glass interposer, or an organic interposer. In this regard, the first dielectric layer **418***a* may be a glass layer, a semiconductor layer (e.g., a silicon layer), or a layer of a polymer material, and the second dielectric layer **418***b* may be another polymer material. For example, the first dielectric layer **418***a* may be silicon substrate and the second dielectric layer may be a polymer material such as polyimide (PI), benzocyclobutene (BCB), or polybenzo-bisoxazole (PBO). In such an embodiment, the first electrical interconnects **416***a* may be formed within the first dielectric layer **418***a* by performing semiconductor device processing operations. For example, processing operations may include patterning and etching the first dielectric layer **418***a* followed by deposition of an electrically conducting material (e.g., Al, Cu, etc.) to form the first electrical interconnects **416***a*.

[0080] According to some embodiments, the second electrical interconnects **416***b* may be formed within the second dielectric layer **418***b* as redistribution interconnects. In this regard, the second dielectric material **418***b* may be formed as sequentially deposited layers of a polymer material such as PI, BCB, PBO, etc. Each of the sequentially deposited layers may then be patterned and an electrically conducting material (e.g., Ti, Cu, Ni, Al) may be sequentially deposited (e.g., by electroplating) to form the second electrical interconnects **416***b* within the second dielectric layer **418***b*. In some embodiments, the second electrical interconnects **416***b* may have a fan-out configuration (not shown).

[0081] FIG. **5**A is a top view of a further electronic/photonic package **500** including a photonic interconnect die **502**, and FIG. **5**B is a vertical cross-sectional view of the electronic/photonic package **500** of FIG. **5**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **5**B is indicated by the cross-section B-B′ in FIG. **5**A. The electronic/photonic package **500** may include a plurality of first EICs **404***a* and a plurality of second EICs **404***b*. Each of the plurality of first EICs **404***a* may provide a first functionality and each of the plurality of second EICs **404***b* may provide a second functionality. For example, the plurality of first EICs **404***a* may be CPU dies, GPU dies, application specific integrated circuits (ASICs), etc., which may provide a functionality to perform computational logic operations. In various embodiments, the plurality of second EICs **404***b* may be memory (e.g., high-bandwidth memory (HBM)) dies that may provide a data storage functionality.

[0082] Each of the plurality of first EICs **404***a* and the plurality of second EICs **404***b* may be attached to, and electrically coupled to, an electronic interposer **408**. The electronic interposer **408** may, in turn, be attached to, and electrically coupled to, a package substrate **410**. The electronic interposer **408** may provide electrical connections between the plurality of first EICs **404***a* and the plurality of second EICs **404***b*. As such, the electronic interposer **408** may provide power to the various components of the electronic/photonic package **500** and may further provide electrical signal pathways between the components of the electronic/photonic package **500**. The package substrate **410** may allow the electronic/photonic package **500** to be connected to other system components such as a printed circuit board (PCB) (not shown).

[0083] To reduce signal delay and ohmic loss, the electronic/photonic package **500** may further include photonic components that may provide optical/photonic signaling functionality. For example, the electronic/photonic package **500** may include a first optical engine **402***a* and a second optical engine **402***b*. Each of the first optical engine **402***a* and the second optical engine **402***b* may be configured to convert electrical signals into photonic signals and vice versa, as described in greater detail with reference to FIG. **4**, above.

[0084] As shown in FIG. **5**B, in addition to the plurality of first EICs **404***a* and plurality of second EICs **404***b*, the first optical engine **402***a* and the second optical engine **402***b* may also be attached to, and electrically coupled to, the electronic interposer **408**. In this regard, the first optical engine **402***a* and the second optical engine **402***b* may each be attached to the electronic interposer **408** with a plurality of first solder material portions **407***a*. The electronic interposer **408** may, in turn, be attached to, and electrically coupled to, the package substrate **410** with a plurality of second solder material portions **407***b*. As such, each of first optical engine **402***a* and the second optical engine **402***b* may receive power from electrical connections provided by the electronic interposer **408**.

[0085] Each of the first optical engine **402***a* and the second optical engine **402***b* may further receive electrical signals from, and may transmit electrical signals to, the electronic interposer **408**. In this regard, the first optical engine **402***a* and the second optical engine **402***b* may communicate with one another and with other components of the electronic/photonic package **500** by way of electrical connections provided by the electronic interposer **408**. For example, one or more of the plurality of first EICs **404***a* may provide electrical control signals to the first optical engine **402***a* and the second optical engine **402***b*.

[0086] Certain electrical signals received by the first optical engine **402***a* may be converted to first photonic signals **310***a* which may be provided to first photonic signal pathways **514***a* within the first optical engine **402***a*. The first photonic signals **310***a* may propagate within the first photonic signal pathways **514***a* and may be coupled into second photonic signal pathways **514***b* within the second optical engine **402***b* through the photonic interconnect die **502**.
[0087] In this regard, the photonic interconnect die **502** may include a dielectric waveguide **104** that may be photonically coupled to a first photonic coupler **112***a* formed at a first end of the dielectric waveguide **104** and a second photonic coupler **112***b* formed at a second end of the dielectric waveguide **104**. The first photonic coupler **112***a* and the second photonic coupler **112***b* may each couple photonic signals into and out of the photonic interconnect die **502** such that a photonic signal pathway connects the first photonic coupler **112***a*, the dielectric waveguide **104**, and the second photonic coupler **112***b*. As shown in FIG. **5**B, the photonic interconnect die **502** may include a substrate **520** with a cladding portion **522** formed over the substrate. The dielectric waveguide **104** may further include a core portion **524** formed within the cladding portion **522**.
[0088] Each of the cladding portion **522** and the core portion **524** may be formed of dielectric materials such that a first index of refraction of the core portion **524** is greater than a second index of refraction of the cladding portion **522**. For example, in some embodiments, the substrate **520** may be a semiconductor such as silicon, the cladding portion **522** may be silicon dioxide, and the core portion may be formed of silicon. Such a structure may be formed by patterning a silicon-on-insulator structure, as described with reference to FIGS. **3**A to **3**D, above. In other embodiments, both the cladding portion **522** and the core portion **524** may be formed of polymer materials, as described in greater detail with reference to FIGS. **14**A to **15**C, below. In a further example, the core portion **524** may be formed by performing a laser-writing operation on a photosensitive polymer material, as described in greater detail with reference to FIGS. **19**A to **19**D, below. In this regard, a micro-structure of the polymer material may be altered (e.g., a refractive index may be increased) in a localized region in response to absorption of laser radiation during the laser-writing operation. The irradiated localized region may then serve as the core portion **524** while a surrounding un-irradiated portion may serve as the cladding portion **522**.
[0089] As shown in FIG. **5**B, each of the first photonic coupler **112***a* and the second photonic coupler **112***b* may be configured as angled reflectors **306***c* that convert vertically propagating photonic signals into horizontally propagating signals and vice versa. As such, a first photonic signal **310***a* propagating vertically along the first photonic signal pathway **514***a* may enter the photonic interconnect die **502** from the first optical engine **402***a*. Once received by the photonic interconnect die **502** from the first optical engine **402***a*, the first photonic signal **310***a* may be converted by the first photonic coupler **112***a* into a horizontally propagating photonic signal **310***c* propagating in the core portion **524** of the dielectric waveguide **104**. In turn, the horizontally propagating photonic signal **310***c* may be received from the dielectric waveguide **104** and may be converted by the second photonic coupler **112***b* into a second vertically propagating photonic signal **310***b* that may be provided to the second optical engine **402***b*. The received second vertically propagating photonic signal **310***b* may then propagate vertically along the second photonic signal pathways **514***b* within the second optical engine **402***b*. Various other types of photonic couplers may be provided in other embodiments. For example, grating couplers **306***a* (e.g., see FIGS. **3**C and **18**C) and edge couplers **306***b* (e.g., see FIGS. **3**D and **12**) and may be provided in other embodiments as described in greater detail, below.
[0090] As further shown in FIG. **5**B, each of the first optical engine **402***a* and the second optical engine **402***b* may be formed as separate dies and may be attached to, and electrically coupled to, the electronic interposer **408**. The photonic interconnect die **502** may also be formed as a separate die and may be attached to the first optical engine **402***a* and the second optical engine **402***b* as shown, for example, in FIG. **5**B. In this regard, a first portion **526***a* of the photonic interconnect die **502** may be mechanically and photonically coupled to the first optical engine **402***a* and a second

portion **526***b* of the photonic interconnect die **502** may be mechanically and photonically coupled to the second optical engine **402***b*. As described in greater detail with reference to FIG. **7**, below, the photonic interconnect die **502** may be coupled to the first optical engine **402***a* and the second optical engine **402***b* using an optical adhesive (**702***a*, **702***b*) that may be transparent to photonic signals.

[0091] As shown in FIGS. **5**A and **5**B, the photonic interconnect die **502** may be configured as an intra-package photonic coupler. In this regard, the photonic interconnect die **502** may couple photonic components (**402***a*, **402***b*) within a package **500** formed on a single electronic interposer **408**. In other embodiments, a photonic interconnect die **502** may couple components formed on two or more different interposers, as described in greater detail with reference to FIGS. **8**A and **8**B, below. As such, the photonic interconnect die **502** may be configured as an inter-package photonic coupler in further embodiments.

[0092] FIG. **6**A is a top view of a photonic interconnect die **502** that photonically connects a first photonic component (**200***a*, **402***a*) and a second photonic component (**200***b*, **402***b*), according to various embodiments. In this regard, the first photonic component may be one of a first PIC **200***a* or a first optical engine **402***a*, and the second photonic component may be one of a second PIC **200***b* or a second optical engine **402***b*. An example embodiment in which the photonic interconnect die **502** couples a first optical engine **402***a* to a second optical engine **402***b* is described in greater detail with reference to FIGS. **5**A and **5**B, above. In a further embodiment, a first optical engine **402***a* may be coupled to a PIC **200**, as described in greater detail with reference to FIGS. **9**A, **9**B, and **10**, below.

[0093] As shown in FIG. **6**A, the photonic interconnect die **502** may include a plurality of dielectric waveguides **104**. Each of the dielectric waveguides may include a core portion **524** forming within a surrounding cladding portion **522**. Each of the plurality of dielectric waveguides **104** may be photonically coupled to a first photonic coupler **112***a* and a second photonic coupler **112***b*. In this example embodiment, each of the plurality of dielectric waveguides **104** may share a common first photonic coupler **112***a* and a common second photonic coupler **112***b*. Each of the common first photonic coupler **112***a* and the common second photonic coupler **112***b* may be formed as an angled reflector **306***c* (e.g., see FIG. **5**B). Other embodiments may include other types of photonic couplers (**306***a*, **306***b*), as described in greater detail below

[0094] In contrast to the common first photonic coupler **112***a* and the common second photonic coupler **112***b*, in other embodiments, each of the core portions **524** may be coupled to a respective individual first photonic coupler **112***a* (not shown) and a respective individual second photonic coupler **112***b* (not shown). For example, each of the core portions **524** may be coupled to respective angled reflectors **306***c* that may be spatially separated from one another. Alternatively, each of the core portions **524** may be coupled to respective grating couplers **306***a*. The formation of angled reflectors **306***c* is described in greater detail with reference to FIGS. **16**A to **16**H, below, and the formation of grating couplers **306***a* is described in greater detail with reference to FIGS. **18**A to **18**C, below. As mentioned with reference to FIG. **5**B, above, the photonic interconnect die **502** may include a substrate **520** (not shown in FIG. **6**A) on which the cladding portion **522** may be formed, as described in greater detail with reference to FIGS. **14**A to **15**C, below. While the photonic interconnect die **502** of FIG. **6**A may couple two photonic components, other embodiments may couple three, four, etc., photonic components, as described in greater detail with reference to FIG. **6**B, below.

[0095] FIG. **6**B is a top view of a photonic interconnect die **502** that photonically connects a first photonic component (**200***a*, **402***a*), a second photonic component (**200***b*, **402***b*), a third photonic component (**200***c*, **402***b*), and a fourth photonic component (**200***d*, **402***d*), according to various embodiments. In this regard, various combinations of PICs (**200***a*, **200***b*, **200***c*, **200***d*) and optical engines (**402***a*, **402***b*, **402***c*, **402***d*) may be photonically coupled. As shown, the photonic interconnect die **502** may include a first plurality of dielectric waveguides **104***a*, a second plurality of dielectric waveguides **104***b*, a third plurality of dielectric waveguides **104***c*, a fourth plurality of dielectric waveguides **104***d*, and a fifth plurality of dielectric waveguides **104***e*.

[0096] In the example embodiment of FIG. **6**B, the first plurality of dielectric waveguides **104***a* may photonically couple the first photonic component (**200***a*, **402***a*) and the second photonic component (**200***b*, **402***b*); the second plurality of dielectric waveguides **104***b* may photonically couple the third photonic component (**200***c*, **402***c*) and the fourth photonic component (**200***d*, **402***d*); the third plurality of dielectric waveguides **104***c* may photonically couple the first photonic component (**200***a*, **402***a*) and the third photonic component (**200***c*, **402***c*); the fourth plurality of dielectric waveguides **104***d* may photonically couple the second photonic component (**200***b*, **402***b*) and the fourth photonic component (**200***d*, **402***d*); and the fifth plurality of dielectric waveguides **104***e* may photonically couple the first photonic component (**200***a*, **402***a*) and the fourth photonic component (**200***d*, **402***d*).

[0097] Each plurality of dielectric waveguides (**104***a*, **104***b*, **104***c*, **104***d*, **104***e*) may include a plurality of core portions **524** formed within a surrounding cladding portion **522**. Each plurality of dielectric waveguides (**104***a*, **104***b*, **104***c*, **104***d*, **104***e*) may also be photonically coupled to respective first photonic couplers **112***a* and second photonic couplers **112***b*. The photonic couplers (**112***a*, **112***b*) may include grating couplers **306***a*, edge couplers **306***b*, and angled reflectors **306***c* in various embodiments (e.g., see FIGS. **3**C, **3**D, and **5**B). As shown in FIG. **6**B, the pluralities of dielectric waveguides (**104***a*, **104***b*, **104***c*, **104***d*, **104***e*) may be configured in various ways including straight dielectric waveguides (**104***a*, **104***b*, **104***c*, **104***d*) as well as curved dielectric waveguides **104***e*. While the photonic interconnect die **502** of FIG. **6**B may couple four photonic components, other embodiments may couple three, five, six, etc., photonic components, in other embodiments.

[0098] FIG. **7** is a vertical cross-sectional view of a further electronic/photonic package **700** including a photonic interconnect die **502**, according to various embodiments. As shown, the electronic/photonic package **700** may include a first optical component (e.g., first optical engine **402***a*) and a second optical component (e.g., second optical engine **402***b*). As shown, a first portion **526***a* of photonic interconnect die **502** may mechanically and photonically coupled to the first optical engine **402***a* and a second portion **526***b* of photonic interconnect die **502** may be mechanically and photonically coupled to the second optical engine **402***b*. In this regard, a first optical adhesive layer **702***a* may connect the first portion **526***a* of photonic interconnect die **502** to a first integrated optics section **412***a* of the first optical engine **402***a*, and a second optical adhesive layer **702***b* may connect the second portion **526***b* of photonic interconnect die **502** to a second integrated optics section **412***b* of the second optical engine **402***b*.

[0099] As shown, the first optical adhesive layer **702***a* and the second optical adhesive layer **702***b* may have different thicknesses to accommodate different sizes and placement/alignment errors of optical components (**402***a*, **402***b*) to be joined by the photonic interconnect die **502**. In this regard, the first optical adhesive layer **702***a* may be thicker than the second optical adhesive layer **702***b*. For example, the first optical engine **402***a* may have a smaller thickness than the second optical engine **402***b*.

[0100] Alternatively, the first optical engine **402***a* and the second optical engine **402***b* may have a common thickness but may have a positioning tolerance that allows a certain height difference due to variations in thicknesses of the first solder material portions **407***a* used to bond the optical components (**402***a*, **402***b*) to the electronic interposer **408**.

[0101] The optical adhesive layers (**702***a*, **702***b*) may be chosen to be transparent and to provide a secure mechanical connection between the photonic interconnect die **502** and the optical components (**402***a*, **402**). According to an embodiment, the optical adhesive layer (**702***a*, **702***b*) may include an adhesive that is similar to materials that may be used to secure optical fibers within connectors, splices, and other components in fiber optic systems. As such, the optical adhesive layers (**702***a*, **702***b*) may include a material that has optical clarity, low shrinkage, and good adhesion properties to ensure efficient light transmission between the photonic interconnect die **502** and the optical components (**402***a*, **402**).

[0102] In some embodiments, the optical adhesive layers (**702***a*, **702***b*) may include an optical epoxy, which may be a two-part adhesive that includes a resin and a hardener. When mixed in the correct proportions and applied between the optical components (**402***a*, **402***b*) and the photonic interconnect die **502**, the optical adhesive layers (**702***a*, **702***b*) may undergo a chemical reaction to cure and form a solid, transparent bond. The cured optical adhesive layers (**702***a*, **702***b*) may thereby form a mechanical connection that maintains an alignment and stability of the optical components (**402***a*, **402***b*) and the photonic interconnect die **502**, minimizing signal loss and ensuring reliable data transmission between the photonic interconnect die **502** and the optical components (**402***a*, **402**).

[0103] Chemical compositions of optical adhesives, or optical epoxies, may vary based on the specific requirements and properties desired for different applications. According to certain embodiments, optical epoxies may include an epoxy resin, a curing agent or hardener, filler materials, plasticizers, adhesion promoters, UV stabilizers, and optical transparency agents. The epoxy resin may provide the primary structure and bonding capability. Such resins may be polymer materials formed by the reaction of an epoxide resin with a curing agent or hardener. Example epoxy resins include bisphenol A (DGEBA) and bisphenol F. The curing agent or hardener component may initiate a polymerization reaction with the epoxy resin, leading to hardening or curing of the adhesive. Amines may be used as curing agents in optical epoxies.

[0104] Fillers may be added to improve mechanical and thermal properties of the optical adhesive layers (**702***a*, **702***b*). In some embodiments, fillers may also enhance the optical properties of the optical adhesive layers (**702***a*, **702***b*). Example fillers may include silica or other fine particles such as carbon/graphite particles. Plasticizers may be added to improve flexibility and to reduce brittleness in the cured adhesive layers (**702***a*, **702***b*). Adhesion promoters may include compounds added to enhance bonding properties to specific materials, such as glass or metal surfaces. According to some embodiments, optical epoxies used in applications that may be exposed to UV light may contain UV stabilizers to prevent degradation of the optical adhesive layers (**702***a*, **702***b*) that may be caused by ultraviolet radiation. According to some embodiments, the optical adhesive layers (**702***a*, **702***b*) may include optical transparency agents to maintain or enhance optical clarity. These agents may be used to ensure that the cured optical adhesive layers (**702***a*, **702***b*) have low optical absorption and thereby do not introduce loss of transmitted photonic signals.

[0105] FIG. **8**A is a top view of a further electronic/photonic package **800** including a first photonic interconnect die **502***a*, a second photonic interconnect die **502***b*, and a third photonic interconnect die **502***c*, and FIG. **8**B is a vertical cross-sectional view of the electronic/photonic package of FIG. **8**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **8**B is indicated by the cross-section B-B′ in FIG. **8**A. As shown in FIG. **8**A, the electronic/photonic package **800** may include a plurality of first EICs **404***a* and a plurality of second EICs **404***b*. Each of the plurality of first EICs **404***a* may provide a first functionality and each of the plurality of second EICs **404***b* may provide a second functionality. For example, the plurality of first EICs **404***a* may be CPU dies, GPU dies, application specific integrated circuits (ASICs), etc., which may provide a functionality to perform computational logic operations. In various embodiments, the plurality of second EICs **404***b* may be memory (e.g., high-bandwidth memory (HBM)) dies that may provide a data storage functionality.

[0106] The electronic/photonic package **800** may further include a first electronic interposer **408***a* and a second electronic interposer **408***b*. Some of the plurality of first EICs **404***a* and the plurality of second EICs **404***b* may be attached to, and electrically coupled to, the first electronic interposer **408***a* and other ones of the plurality of first EICs **404***a* and the plurality of second EICs **404***b* may be attached to, and electrically coupled to, the second electronic interposer **408***b*. In turn, each of the first electronic interposer **408***a* and the second electronic interposer **408***b* may be attached to, and electrically coupled to, a package substrate **410**. The package substrate **410** may allow the electronic/photonic package **800** to be connected to other system components such as a printed circuit board (PCB) (not shown). The first electrical interposer **408***a* and the second electrical interposer **408***b* may provide electrical connections to various ones in the plurality of first EICs **404***a* and the plurality of second EICs **404***b*.

[0107] As with other embodiments described above, the electronic/photonic package **800** may further include photonic components that may provide optical/photonic signaling functionality. For example, the electronic/photonic package **800** may include a first optical engine **402***a*, a second optical engine **402***b*, a third optical engine **402***c*, and a fourth optical engine **402***d*. Each of the first optical engine **402***a*, the second optical engine **402***b*, the third optical engine **402***c*, and the fourth optical engine **402***d* may be configured to convert electrical signals into photonic signals and vice versa, as described in greater detail with reference to FIGS. **4**, **5**A, and **5**B, above. The first optical engine **402***a* and the second optical engine **402***b* may be attached to, and electrically coupled to, the first electronic interposer **408***a*. Similarly, the third optical engine **402***c* and the fourth optical engine **402***d* may be attached to, and electrically coupled to, the second electronic interposer **408***b*.

[0108] Each of the first optical engine **402***a* and the second optical engine **402***b* may receive electrical signals from, and may transmit electrical signals to, the first electronic interposer **408***a*, and each of the third optical engine **402***c* and the fourth optical engine **402***d* may receive electrical signals from, and may transmit electrical signals to, the second electronic interposer **408***b*. The first photonic interconnect die **502***a* may photonically couple the first optical engine **402***a* and the second optical engine **402***b*, and as such, may be configured as an intra-package photonic coupler. Similarly, the third photonic interconnect die **502***c* may also be configured as an intra-package photonic coupler that photonically couples the third optical engine **402***c* with the fourth optical engine **402***d*. In contrast, the second photonic interconnect die **502***a* may be configured as an inter-package photonic coupler by photonically coupling the second optical engine **402***b* with the third optical engine **402***c*. In this regard, signals may be efficiently routed between devices attached to the first electrical interposer **408***a* and devices attached to the second electrical interposer **408***b*.

[0109] As shown in FIGS. **8**A and **8**B, the electronic/photonic package **800** may further include a fiber array unit (FAU) **802**. The FAU **802** may couple photonic signals into and out of the fourth optical engine **402***b*. As shown in FIG. **8**B, the FAU **802** may include a housing **804** having a curved reflector surface **806** and a fiber coupler section **808**. A fiber optic cable **810** may be mechanically and photonically coupled to the fiber coupler section **804**. As such, optical fibers **204** within the fiber optic cable **810** may be photonically coupled to the housing **804** such that a photonic signal **310** propagating vertically within the fourth optical engine **402***d* may be reflected by the curved reflector surface **806** into the fiber coupler section **804**. The photonic signal **310** received from the fourth optical engine **402***d* may then be transmitted into one or more of the optical fibers **204** within the fiber optic cable **810**. Similarly, photonic signals **310** received from the fiber optic cable **810** may be reflected by the curved reflector surface **806** into the fourth optical engine **402***d*.

[0110] FIG. **8**C is a three-dimensional perspective view showing details of the FAU **802** of FIG. **8**B, according to various embodiments. As shown, the housing **804** may include a curved reflector surface **806**. Optical fibers **204** may be placed in respective groves **812** formed in the housing **804** in the fiber coupler section **808**. An optical adhesive **702** may be formed between ends of the optical fibers **204** and a coupling surface **814** of the housing **804**. The housing **804** may be formed of a polymer material that may be patterned and etched using semiconductor manufacturing techniques. For example, the curved reflector surface **806** may be formed by patterning and etching the polymer material of the housing **804** using multi-tone masks and low contrast photoresists, according to various embodiments.

[0111] The dashed region of the fiber coupler section **808** may also be formed by removing a portion of the polymer material forming the housing **804** using patterning and etching techniques. Similarly, the groves **812** that secure the optical fibers may be generated by performing an etching process. As shown in FIG. **8**B, the optical fibers **204** may be provided in a fiber optic cable **810** (omitted in FIG. **8**C for clarity). Once the optical fibers **204** are secured within the groves of the housing **804**, an additional polymer material (not shown) may be formed over the optical fibers **204** within a volume indicated by the dashed region of the fiber coupler section **808**. The use of a single angled reflector surface **806** for a plurality of optical fibers **204** allows multiple optical fibers **204** to be closely spaced without a need for tight alignment tolerances that may otherwise be required if each optical fiber **204** was to be aligned with its own respective reflector.

[0112] As further shown in FIG. **8**C, the grooves **812** may be formed to have internal surfaces (**813***a*, **313***b*) that may be angled relative to a plane **815** parallel to a surface **817** of the housing **804**. In this regard, a first surface **813***a* may subtend a first angle θ.sub.1 relative to the plane **815**. A second surface **813***b* may have a similar angle (not shown) relative to the plane **815**. A fixed diameter **819** optical fiber **204** may be seated within the groove **812** to a depth H.sub.1 relative to the surface **817** of the housing **804**, such that the depth H.sub.1 is a geometric function of the first angle θ.sub.1. Alternatively, in a further embodiment, a groove may be wider and may therefore be characterized by a second angle θ.sub.2 (with θ.sub.2>θ.sub.1) measured relative to the plane **815**. In this example, the wider groove may accommodate a fixed diameter **819** optical fiber **204** to a depth H.sub.2 that is greater than the depth H.sub.1 of the first groove. The depth H.sub.2 is given by the same geometric function of the second angle θ.sub.2 as the function that determines the first depth H.sub.1 as a function of the first angle θ.sub.1. The first depth H.sub.1 and the second depth H.sub.2 measure a vertical position of the optical fiber **812** relative to the surface **817**. Thus, the vertical positioning of the optical fiber **204** may be designed to have a pre-determined depth by forming the groove **812** to having internal surfaces (**813***a*, **813***b*) that subtend corresponding pre-determined angles relative to the horizontal plane **815**.

[0113] Similarly, as further shown in FIG. **8**C, a curvature of the curved reflector surface **806** may be chosen to provide a certain pre-determined reflectivity of a photonic signal provided by the optical fibers **204**. In this regard, an incident photonic signal **821***a* may impinge on the curved reflector **806** and may be reflected to thereby generate a reflected photonic signal **821***b*. As shown, the reflected photonic signal **821***b* may propagate away from the curved reflector surface **806** at an angle **823** relative to the incident photonic signal **821***a*. The reflected angle **823** may depend on an angle of incidence (not explicitly shown) between the incident photonic signal **821***a* and the curved reflector surface **806**. The reflected angle **823** may further depend on a curvature (i.e., a radius of curvature) of the curved reflector surface **806**. Thus, a pre-determined refection characteristic (e.g., a reflection angle **823**) may be determined by choosing the curvature of the curved reflector surface **806** to have a certain pre-determined value (i.e., local radius of curvature). As such, the coupling between the optical fibers **204** and the FAU **802** may be determined by the curvature of the curved reflector surface **806**, as well as by adjusting a vertical position of the optical fibers **204**, as described above.

[0114] In some embodiments, the curved reflector surface **806** may have a single curvature (e.g., labeled “Curvature 1”) or may have a variable curvature. A single curvature means a single value of a radius of curvature that characterizes the entire surface. In other embodiments, the surface may have a curvature (i.e., radius of curvature) that varies along the surface. In one example, a surface may a have a first portion having a first radius of curvature (e.g., labeled “Curvature 1”) and a second portion having a second radius of curvature (e.g., labeled “Curvature 2”), according to further embodiments. In still-further embodiments, the curved reflector surface **806** may have a radius of curvature that varies continuously (not shown) as a function of position on the curved reflector surface **806**. In other embodiments, the radius of curvature may be a piecewise-smooth function of position (not shown) on the curved reflector surface **806** (i.e., may include N segments each corresponding to N different respective curvatures, where N is an integer).

[0115] FIG. **9**A is a top view of a further electronic/photonic package **900** including a plurality of photonic interconnect dies (**502***a*, **502***b*, **502***c*, **502***d*, **502***e*, **502***f*, **502***g*, **502***h*), according to various embodiments. The electronic/photonic package **900** may include a plurality of first EICs **404***a* and a plurality of second EICs **404***b*. Each of the plurality of first EICs **404***a* may provide a first functionality (e.g., logic operations) and each of the plurality of second EICs **404***b* may provide a second functionality (e.g., data storage operations).

[0116] Each of the plurality of first EICs **404***a* and the plurality of second EICs **404***b* may be attached to, and electrically coupled to, an electronic interposer **408**. The electronic interposer **408** may, in turn, be attached to, and electrically coupled to, a package substrate **410**. The electronic interposer **408** may provide electrical connections between the plurality of first EICs **404***a* and the plurality of second EICs **404***b*. As such, the electronic interposer **408** may provide power to the various components of the electronic/photonic package **900** and may further provide electrical signal pathways between the components of the electronic/photonic package **900**. The package substrate **410** may allow the electronic/photonic package **500** to be connected to other system components such as a printed circuit board (PCB) (not shown).

[0117] As shown in FIG. **9**A, some of the first EICs **404***a* may be attached to, and electrically coupled to, an active interposer **902**. The active interposer may provide electrical connections between neighboring first EIC **404***a* and may further include active circuit components. For example, the active interposer **902** may include transistors that may provide additional logic processing operations and/or data storage functionality. Some of the second EICs **404***b* may further be co-packaged with one or more first EICs **404***a* to form a stacked structure **904**. In this regard, the stacked structure **904** may include a second EIC **404***b* stacked on top of a first EIC **404***a*. For example, a the second EIC **404***b* may be bonded to the first EIC **404***a* in the stacked structure **904** using hybrid bonding structures (not shown). In this regard, the first EIC **404***a* may provide control circuitry for the second EIC **404***b*, which may provide a data storage functionality.

[0118] To reduce signal delay and ohmic loss, the electronic/photonic package **900** may further include photonic components that may provide optical/photonic signaling functionality. For example, the electronic/photonic package **900** may include a plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*, **200***e*, **200***f*, **200***g*). Signal pathways within the electronic/photonic package **900** may thus include both electronic signal pathways (e.g., see solid arrows) as well as photonic signal pathways (e.g., see dot-dashed arrows). As shown in FIG. **9**A, electronic signal pathways may exist between neighboring first EICs **404***a*, between first EICs **404***a* and second EICs **404***b*, between first EICs **404***a* and one or more PICs (**200***a*, **200***b*, **200***c*, **200***d*, **200***e*, **200***f*, **200***g*) and between second EICs **404***b* and one or more PICs (**200***a*, **200***b*, **200***c*, **200***d*, **200***e*, **200***f*, **200***g*).

[0119] In addition to reducing ohmic loss by providing photonic signaling pathways, one or more of the plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*, **200***e*, **200***f*, **200***g*) may provide a functionality to perform (classical or quantum) logic operations directly on photonic signals. As such, various ones of the plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*, **200***e*, **200***f*, **200***g*) may provide respective functionalities and may have respective structural configurations. For example, some of the PICs (**200***d*, **200***e*, **200***f*, **200***g*) may be smaller than other PICs (**200***a*, **200***b*, **200***c*) and may have more limited functionality than that of the larger PICs (**200***a*, **200***b*, **200***c*). For example, the smaller PICs (**200***d*, **200***e*, **200***f*, **200***g*) may be configured as photonic transceivers, as described with reference to FIG. **2**, above.

[0120] Alternatively, the larger PICs (**200***a*, **200***b*, **200***c*) may be configured as monolithic electronic/photonic devices that may perform a variety of functions including photonic signal generation and photonic signal processing. For example, the monolithic PICs (**200***a*, **200***b*, **200***c*) may receive electrical signals from neighboring EICs (**404***a*, **404***b*) as well as photonic signals from neighboring PICs (**200***d*, **200***e*, **200***f*, **200***g*) and may perform a variety of processing operations on the received signals. The electronic/photonic package **900** may further include an FAU **802** that may be configured to transmit and receive photonic signals. As shown, the FAU **802** may provide a photonic interface between one of the PICs (e.g., PIC **200***b*) and with external photonic circuits (not shown). The FAU **802** may be configured similarly to that of the FAU **802** described above with reference to FIGS. **8**B and **8**C and may provide similar photonic signaling functionality.

[0121] FIG. **9**B is a vertical cross-sectional view of a portion of the

[0122] electronic/photonic package **900** of FIG. **9**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **9**B is indicated by the cross-section B-B′ in FIG. **9**A. As shown, the portion of the electronic/photonic package **900** of FIG. **9**A includes a first photonic interconnect die **502***a* and a second photonic interconnect die **502***b*. The first photonic interconnect die **502***a* may photonically connect a first PIC **200***a* with a second PIC **200***d*. Similarly, the second photonic interconnect die **502***b* may photonically connect the second PIC **200***d* with a third PIC **200***c*. As described above, the various PICs (**200***a*, **200***c*, **200***d*) may have respective configurations and may provide respective functionalities. For example, the second PIC **200***c* may have relatively limited functionality and may, for example, be configured as a photonic transceiver, as described above with reference to FIG. **2**. In contrast, the first PIC **200***a* and the third PIC **200***c* may be formed as monolith structures that may provide more complex functionality including. for example, computing and logic operations on photonic data.

[0123] FIG. **10**A is a top view of a further electronic/photonic package **1000** including a plurality of photonic interconnect dies (**502***a*, **502***b*, **502***c*, **502***d*, **502***e*, **502***f*, **502***g*, **502***h*), and FIG. **10**B is a vertical cross-sectional view of a portion of the electronic/photonic package **1000** of FIG. **10**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **10**B is indicated by the cross-section B-B′ in FIG. **10**A. The electronic/photonic package **1000** may be similar to the electronic/photonic package **900** of FIGS. **9**A and **9**B. In this regard, the electronic/photonic package **1000** may include a plurality of first EICs **404***a* and a plurality of second EICs **404***b*. Each of the plurality of first EICs **404***a* may provide a first functionality (e.g., logic operations) and each of the plurality of second EICs **404***b* may provide a second functionality (e.g., data storage operations).

[0124] Each of the plurality of first EICs **404***a* and the plurality of second EICs **404***b* may be attached to, and electrically coupled to, an electronic interposer **408**. The electronic interposer **408** may, in turn, be attached to, and electrically coupled to, a package substrate **410**. The electronic interposer **408** may provide electrical connections between the plurality of first EICs **404***a* and the plurality of second EICs **404***b*. As such, the electronic interposer **408** may provide power to the various components of the electronic/photonic package **900** and may further provide electrical signal pathways between the components of the electronic/photonic package **900**. The package substrate **410** may allow the electronic/photonic package **500** to be connected to other system components such as a printed circuit board (PCB) (not shown).

[0125] In contrast to the electronic/photonic package **900** of FIGS. **9**A and **9**B, the electronic/photonic package **1000** may omit the active interposer **902** and the stacked structure **904**. As such, all of the first EICs **404***a* and the second EICs **404***b* may be attached, and electrically coupled, directly to the electronic interposer **408**. As such, the electronic/photonic package **1000** may be simpler in terms of electrical connections formed between the various EICs (**404***a*, **404***b*) and the electrical interposer **408**. The relative simplicity of the electronic/photonic package **900** of FIGS. **9**A and **9**B may be advantageous for certain applications.

[0126] As with the electronic/photonic package **900** of FIGS. **9**A and **9**B, the electronic/photonic package **1000** of FIGS. **10**A and **10**B may further include photonic components that may provide optical/photonic signaling functionality. For example, the electronic/photonic package **1000** may include a plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*) and optical engines (**402***a*, **402***b*, **402***c*). Signal pathways within the electronic/photonic package **1000** may thus include both electronic signal pathways (e.g., see solid arrows) as well as photonic signal pathways (e.g., see dot-dashed arrows). As shown in FIG. **10**A, electronic signal pathways may exist between neighboring first EICs **404***a*, between first EICs **404***a* and second EICs **404***b*, between first EICs **404***a* and one or more optical engines (**402***a*, **402***b*, **402***c*) and between second EICs **404***b* and one or more optical engines (**402***a*, **402***b*, **402***c*).

[0127] As with the electronic/photonic package **900** of FIGS. **9**A and **9**B, one or more of the plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*) may provide a functionality to perform (classical or quantum) logic operations directly on photonic signals, in addition to reducing ohmic loss by providing photonic signaling pathways. As such, various ones of the plurality of PICs (**200***a*, **200***b*, **200***c*, **200***d*) may provide respective functionalities and may have respective structural configurations. For example, some of the PICs (**200***a*, **200***b*, **200***c*, **200***d*) may have more limited functionality than that of other ones of the PICs (**200***a*, **200***b*, **200***c*, **200***d*). For example, some of the PICs (**200***a*, **200***b*, **200***c*, **200***d*) may be configured as photonic transceivers, as described with reference to FIG. **2**, above. Alternatively, other ones of the PICs (**200***a*, **200***b*, **200***c*, **200***d*) may be configured to perform a variety of functions including photonic signal generation and photonic signal processing.

[0128] In contrast to other embodiments described above (e.g., see FIGS. **4** to **5**B), the optical engines (**402***a*, **402***b*, **402***c*) may be larger and may be configured to provide additional functionality to other optical engines described above. In this regard, the optical engines (**402***a*, **402***b*, **402***c*) of FIGS. **10**A and **10**B may be larger, monolithic devices that may provide enhanced functionality. In this regard, the optical engines (**402***a*, **402***b*, **402***c*) of FIGS. **10**A and **10**B may be configured to be electrically connected to a plurality of first EICs **404***a* and second EICs **404***b* in addition to being photonically coupled with two or more PICs. For example, as shown in FIG. **10**A, a first monolithic optical engine **402***a* may be electronically coupled to second EICs **404***b* that are attached to the electronic interposer **408** adjacent to the first monolithic optical engine **402***a*.

[0129] Further, the first monolithic optical engine **402***a* may be photonically coupled to a first PIC **200***a* and a second PIC **200***b*. Thus, according to certain embodiments, photonic logic operations may be performed by the first PIC **200***a* and the second PIC **200***b* and the results of the photonic computations may be provided to the first monolithic optical engine **402***a* by a first photonic interconnect die **502***a* and a second photonic interconnect die **502***b*, respectively. The photonic signals received by the first monolithic optical engine **402***a* may then be converted to corresponding electrical signals that encode data representing the results of the logic operations performed by the first PIC **200***a* and the second PIC **200***b*. In turn, the data may then be provided to one or more data storage devices provided by the second EICs that are adjacent to the first monolithic optical engine **402***a*, which may store the corresponding data.

[0130] The electronic/photonic package **1000** may further include an FAU **802** that may be configured to transmit and receive photonic signals. As shown, the FAU **802** may provide a photonic interface between one of the second monolithic optical engine **402***b* and external photonic circuits (not shown). The FAU **802** may be configured similarly to that of the FAU **802** described above with reference to FIGS. **8**B and **8**C and may provide similar photonic signaling functionality.

[0131] FIG. **10**B is a vertical cross-sectional view of a portion of the electronic/photonic package **1000** of FIG. **10**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **10**B is indicated by the cross-section B-B′ in FIG. **10**A. As shown, the portion of the electronic/photonic package **1000** of FIG. **10**A includes a first photonic interconnect die **502***a* and a second photonic interconnect die **502***b*. The first photonic interconnect die **502***a* may photonically connect the first monolithic optical engine **402***a* with a first PIC **200***a*. Similarly, the second photonic interconnect die **502***b* may photonically connect the first PIC **200***a* with the second monolithic optical engine **402***b*.

[0132] FIG. **11** is a top view of a further electronic/photonic package **1100** including a first component package **1102***a* and a second component package **1102***b*, according to various embodiments. The first component package **1102***a* may include a plurality of first EICs **404***a* attached to a first electronic interposer **408***a* and the second component package **1102***b* may include a plurality of second EICs **404***b*. As shown, the first electronic interposer **408***a* may be attached to a first package substrate **410***a* and the second electronic interposer **408***b* may be attached to a second package substrate **410***b*. The first component package **1102***a* may be optimized to perform computational logic operations and the second package component **1102***b* may be optimized to perform data storage and retrieval operations.

[0133] In this regard, the plurality of first EICs **404***a*, of the first component package **1102***a*, may be CPU dies, GPU dies, application specific integrated circuits (ASICs), etc., and the plurality of second EICs **404***b*, of the second package component **1102***b*, may be memory (e.g., high-bandwidth memory (HBM)) dies that may provide a data storage functionality. As described with reference to FIGS. **9**A and **9**B, one or more of the first EICs **404***a* of the first component package **1102***a* may be attached to an active interposer **902**, and one or more of the second EICs **404***b* of the second package component **1102***b* may be attached to a first EIC **404***a* in a stacked structure **904**. Alternative embodiments may omit the active interposer **902** and the EIC **404***a* of the stacked structure **904**.

[0134] Each of the first component package **1102***a* and the second component package **1102***b* may further include photonic components that may provide optical/photonic signaling functionality. For example, each of the first component package **1102***a* and the second component package **1102***b* may include one or more PICs (**200***a*, **200***b*, **200***c*, **200***d*) and one or more optical engines (**402***a*, **402***b*). As such, each of the first component package **1102***a* and the second component package **1102***b* may include electronic signaling pathways (e.g., see solid arrows) and photonic signaling pathways (e.g., see dot-dashed arrows).

[0135] As described above, the various PICs (**200***a*, **200***b*, **200***c*, **200***d*) may have different configurations and may provide respective functionalities. For example, some of the PICs (**200***a*, **200***b*) may be configured as monolithic PICs, as described in greater detail with reference to FIGS. **9**A and **9**B, while other ones of the PICS (**200***c*, **200***d*) may have a simpler structure and may perform respective functions. Further, as described with reference to FIGS. **10**A and **10**B, one or both of the first component package **1102***a* and the second component package **1102***b* may include one or more monolithic optical engines (**402***a*, **402***b*). As described above, the one or more monolithic optical engines (**402***a*, **402***b*) may be configured to interact with multiple EICs (**404***a*, **404***b*) and may convert electronic signals to photonic signals and vice-versa.

[0136] As further shown in FIG. **11**, the first component package **1102***a* and the second component package **1102***b* may be photonically coupled to one another by a fiber optic cable **810**. In this regard, the first component package **1102***a* may include a first FAU **802***a* that is coupled to a first end of the fiber optic cable **810** and the second component package **1102***b* may include a second FAU **802***b* that may be coupled to a second end of the fiber optic cable **810**. Data may be transferred between the first component package **1102***a* and the second component package **1102***b* by way of the fiber optic cable **810**. As shown, the first FAU **802***a* and the second FAU **802***b* may be photonically coupled to respective monolithic PICs **200***b* or optical engines **402***b*.

[0137] Data generated by the first component package **1102***a* as a result of computational logic processes performed by the first EICs **404***a* may be transferred to the second component package **1102***b* in the form of photonic signals. Such photonic signals received by the second component package **1102***b* may then be converted to electrical signals and may be provided to the second EICs **404***b* for storage as digital data. Similarly, data stored on the second EICs **404***b* of the second component package **1102***b* may be retrieved and sent to the first component package **1102***a* as needed for ongoing logic operations being performed by the first EICs **404***a* of the first component package **1102***a*.

[0138] As indicated by the ellipsis **1104** in FIG. **11**, the first component package **1102***a* and the second component package **1102***b* may represent one repeat unit in an electronic/photonic package **1100** including multiple component packages. As such, a plurality of first component packages **1102***a* may form a pool of computational resources that may perform logic operations. Similarly, a plurality of second component packages **1102***b* may provide a pool of memory resources that may perform data storage and retrieval operations.

[0139] FIG. **12** is a vertical cross-sectional view of a further electronic/photonic package **1200** including a photonic interconnect die **502** having edge couplers **306**_b_, according to various embodiments. As shown, the electronic/photonic package **1200** may include a first optical engine **402**_a_ and a second optical engine **402**_b_. Each of the first optical engine **402**_a_ and the second optical engine **402**_b_ may be attached to, and electrically coupled to, an electronic interposer **408**. The electronic interposer **408** may be further attached to, and electrically coupled to, a package substrate **410**.

[0140] The photonic interconnect die **502** may include a substrate **520** and a cladding layer **522** formed over the substrate **520**. The photonic interconnect die **502** may further include a core portion **524** formed within cladding portion **522** such that that core portion **524** and the cladding portion **522** form a dielectric waveguide **104**. As in other embodiments, described above, the core portion **524** may be chosen to be a dielectric material having a larger index of refraction than that of the cladding portion **522**. As such, photonic signals may preferentially propagate within the core portion **524** of the dielectric waveguide.

[0141] In contrast to embodiments described above (e.g., see FIG. **5**B), however, the photonic interconnect die **502** may be configured such that the photonic signals may be coupled into and out from the photonic interconnect die **502** by way of edge couplers **306**_b_. As described with reference to FIG. **3**D, above, an edge coupler **306**_b_ may include a flat surface of the waveguide **104** such that photonic signals may be coupled into and out of ends of the dielectric waveguide **104**. In some embodiments, the edge couplers **306**_b_ may further include a spot size converter (e.g., see FIGS. **17**A and **17**B) that allows a lateral size of an optical mode to be increased such that photonic signals may be efficiently coupled between a dielectric waveguide **104** and an optical fiber (also not shown).

[0142] As shown in FIG. **12**, the photonic interconnect die **502** may be formed such that the substrate **520** has a first width **1202**_a_ that is larger than a second width **1202**_b_ of dielectric waveguide **104**. As such, the photonic interconnect die **502** may be placed such that respective edges of the substrate **520** may be placed in contact with respective top edges of the first optical engine **402**_a_ and the second optical engine **402**_b_. In this regard, the photonic interconnect die **502** may be mechanically supported but the top edges of the first optical engine **402**_a_ and the second optical engine **402**_b_. As shown, side edges of the core portion **524** and the cladding portion **522** may contact respective side edges of the first optical engine **402**_a_ and the second optical engine **402**_b_. As such, photonic signals may be coupled from the optical engines (**402**_a_, **402**_b_) into the photonic interconnect die **502** and vice-versa using an edge-coupling configuration (i.e., with edge couplers **306**_b_).

[0143] FIG. **13** is a vertical cross-sectional view of a further electronic/photonic package including a photonic interconnect die **502** that couples photonic signals between two hybrid interposers (**408**_a_, **408**_b_), according to various embodiments. As shown, the electronic/photonic package **1300** may include a first optical engine **402**_a_ and a second optical engine **402**_b_. Each of the first optical engine **402**_a_ and the second optical engine **402**_b_ may be attached to, and electrically coupled to, respective interposers (**408**_a_, **408**_b_). Each of the interposers (**408**_a_, **408**_b_) may be further attached to, and electrically coupled to, respective package substrates (**410**_a_, **410**_b_).

[0144] As shown in FIG. **13**, each of the interposers (**408**_a_, **408**_b_) may be configured as hybrid interposers (**408**_a_, **408**_b_) that include both electronic signal pathways and photonic signal pathways. In this regard, each the hybrid interposers (**408**_a_, **408**_b_) may be similar to a PIC **200** (e.g., see FIG. **2** and related description) and may include passive components, such as dielectric waveguides **104**, as well as active components such as photonic sources **102**, detectors **106**, and modulators **108** (e.g., see FIG. **1** and related description). As such, the photonic interconnect die **502** may be attached to respective surfaces of the hybrid interposers (**408**_a_, **408**_b_). In this regard, photonic signals may be routed from the first hybrid interposer **408**_a_, through the photonic interconnect die **502**, and into the second hybrid interposer **408**_b_, and vice-versa.

[0145] FIGS. **14**A to **14**D are vertical cross-sectional views of respective intermediate structures

(**1400***a* to **1400***d*) that may be used to form a photonic interconnect die **502**, according to various embodiments. The intermediate structure **1400***a* of FIG. **14**A may include a substrate **520**, a cladding portion **522** formed over the substrate **520**, and a core material layer **524**L formed over the cladding portion. According to various embodiments, the cladding portion **522** and the core material layer **524**L may each be dielectric materials such that the core material layer **524**L has an index of refraction that is greater than that of the cladding portion **522**. For example, the core material layer **524**L may be a silicon layer and the cladding portion **522** may be silicon oxide. The substrate **520** may be any suitable substrate that may provide mechanical support for the cladding portion **522** and the core material layer **524**L. In alternative embodiments, the cladding portion **522** and the core material layer **524**L may each be polymer materials that may be chosen such that the index of refraction of the core material layer **524**L is greater than that of the cladding material layer **524**L.

[0146] As shown in FIGS. **14**A and **14**B, the intermediate structures **1400***a* and **1400***b* may further include a patterned photoresist **1402** formed over the core material layer **524**. The patterned photoresist **1402** may be formed by depositing a blanket layer of photoresist (not shown) over the core material layer **524**L. The blanket layer of photoresist may then be patterned using lithographic techniques to form the patterned photoresist **1402**. The patterned photoresist **1402** may then be used as an etch mask during an etching process that may be used to etch the core material layer **524**L. As shown in FIG. **14**B, a plurality of core portions **524** may be formed upon performing such an etching process. As shown in FIG. **14**C, the patterned photoresist **1402** may then be removed by ashing or dissolution in a solvent leaving the plurality of core portions **524** formed over the cladding portion **522**.

[0147] As shown in FIG. **14**D, an additional layer of cladding material may then be deposited over the plurality of core portions **524** to extend a thickness of the cladding portion **522**. In this regard, the additional material that is deposited may thereby surround the core portions **524** such that the core portions **524** are formed within the cladding portion. As such, each of the core portions **524** may thereby form core portions **524** of respective dielectric waveguides **104**.

[0148] FIG. **14**E is a top view of a further intermediate structure **1400***e* that may be used to form a photonic interconnect die **502**, according to various embodiments. As shown in FIG. **14**E, each of the core portions **524** may extend along a first direction (i.e., the x-direction). The core portions **524** may further be separated from one another along a second direction (i.e., the y-direction) that is perpendicular to the first direction. As such, each core portion **524**, along with a surrounding part of the cladding portion **522** may form a dielectric waveguide.

[0149] FIGS. **15**A to **15**C are vertical cross-sectional views of respective intermediate structures (**1500***a* to **1500***c*) that may be used to form a photonic interconnect die **502**, according to various embodiments. The intermediate structure **1500***a* of FIG. **15**A may be formed by depositing an additional core material layer **524**L over the intermediate structure **1400***d* of FIG. **14**D followed by forming a patterned photoresist **1402** over the additional core material layer **524**L. In this regard, the patterned photoresist **1402** may be formed by depositing a blanket layer of photoresist (not shown) over the additional core material layer **524**L. The blanket layer of photoresist may then be patterned using lithographic techniques to form the patterned photoresist **1402**. The patterned photoresist **1402** may then be used as an etch mask during an etching process that may be used to etch the core material layer **514**L. As shown in FIG. **15**B, an additional plurality of core portions **524** may be formed upon performing such an etching process. The patterned photoresist **1402** may then be removed by ashing or dissolution in a solvent leaving the plurality of core portions **524** formed over the cladding portion **522**.

[0150] As shown in FIG. **15**C, an additional layer of cladding material may then be deposited over the plurality of core portions **524** to extend a thickness of the cladding portion **522**. In this regard, the additional material that is deposited may thereby surround the additional core portions **524** such that the core portions **524** are formed within the cladding portion **522**. As such, each of the additional core portions **524** may thereby form core portions **524** of respective dielectric waveguides **104**. Thus, a plurality of dielectric waveguides **104** may be formed in a three-dimensional configuration, by repeating the processes described above with reference to FIGS. **14**A to **15**C.

[0151] FIGS. **16**A to **16**C are vertical cross-sectional views of respective intermediate structures (**1600***a* to **1600***c*) that may be used to form a photonic interconnect die **502**, according to various embodiment, and FIG. **16**D is a vertical cross-sectional view of a photonic interconnect die **502** formed by processes described with reference to FIGS. **16**A to **16**C, according to various embodiments. The intermediate structure **1600***a* may include a core portion **524** formed within a cladding portion **522**. As described above with reference to FIGS. **14**A to **15**C, the core portion **524** may be one of a plurality of core portions **524**. As shown in FIG. **16**B, a first angled trench **1602***a* and a second angled trench **1602***b* may be formed in the cladding portion **522**. The first angled trench **1602***a* and the second angled trench **1602***b* may be formed by performing an etch process to etch the cladding portion **522**. In this regard, multi-tone masks may be used to perform an etch process that generates angled surfaces.

[0152] A dielectric material layer **1604**L may then be deposited over the intermediate structure **1600***b* to thereby form the intermediate structure **1600***c* of FIG. **16**C. A planarization process (e.g., chemical mechanical planarization) may then be performed to remove an excess portion of the dielectric material layer **1604**L above a top surface of the cladding portion **522** to thereby form the photonic interconnect die **502** of FIG. **16**D. As shown in FIG. **16**D, remaining portions of the dielectric material layer **1604**L may fill the angled trenches (**1602***a*, **1602***b*) such that a first dielectric window **1604***a* and a second dielectric window **1604***b* may be formed.

[0153] As shown in FIG. **16**D, the photonic interconnect die **502**, may include a substrate **520**, a dielectric waveguide **104**, including a core portion **524** and a cladding portion **522**, formed on the substrate **520**. The photonic interconnect die **502** may further include a first photonic coupler **112***a* formed at a first end of the dielectric waveguide **104** and a second photonic coupler **112***b* formed at a second end of the dielectric waveguide **104**. As shown in FIG. **16**D, the dielectric waveguide **104** may have a planar geometry within the cladding portion **522** such that a surface **1606** of the dielectric waveguide **104** (e.g., a surface **1606** of the core portion **524**) is parallel to a first surface **1608***a*, of the photonic interconnect die **502**. Further, as shown in FIG. **16**D, the first photonic coupler **112***a* and the second photonic coupler **112***b* each couple photonic signals **310** into and out of the photonic interconnect die such that a photonic signal pathway (see dot-dashed lines) connects the first photonic coupler **112***a*, the dielectric waveguide **104**, and the second photonic coupler **112***b*.

[0154] Further, as shown in FIG. **16**D, a first dielectric window **1604***a* may be located at a first position (e.g., left side) on the first surface **1608***a* of the photonic interconnect die **502**, and a second dielectric window **1604***b* may be located at a second position (e.g., right side) on the first surface **1608***a* of the photonic interconnect die **502**. In this regard, the first photonic coupler **112***a* and the second photonic coupler **112***b* each the couple photonic signals **310** into and out of the photonic interconnect die **502** through the first dielectric window **1604***a* and the second dielectric window **1604***b*, respectively.

[0155] As shown in FIG. **16**D, the first photonic coupler **112***a* may be configured as a first angled reflector **306***c* (e.g., see FIG. **5**B) that couples photonic signals **310** into the dielectric waveguide **104** that are received from the first dielectric window **1604***a*. The first photonic coupler **112***a* may further transmit photonic signals through the first dielectric window **1604***a* that are received from the dielectric waveguide **104**. Similarly, the second photonic coupler **112***b* may be configured as a second angled reflector **306***c* (e.g., see FIG. **5**B) that couples photonic signals **310** into the dielectric waveguide **104** that are received from the second dielectric window **1604***b* and transmits photonic signals through the second dielectric window **1604***b* that are received from the dielectric waveguide **104**.

[0156] FIGS. **16**E to **16**H are vertical cross-sectional views of additional photonic interconnect dies **502** including angled reflectors, according to various embodiments. As shown in FIG. **16**E, each angled reflector may further include a reflective coating **1610**. The reflective coating may be chosen to be a metallic material that may be deposited over each of the angled trenches (**1602***a*, **1602***b*) of the intermediate structure **1600***b* of FIG. **16**B prior to deposition of the dielectric material layer **1604**L. Alternatively, as shown in FIG. **16**F, a multi-layer dielectric stack **1612** may be formed over each over each of the angled trenches (**1602***a*, **1602***b*) of the intermediate structure **1600***b* of FIG. **16**B prior to deposition of the dielectric material layer **1604**L.
[0157] The multi-layer dielectric stack **1612** may include thin alternating layers of dielectric materials such that the multi-layer dielectric stack **1612** may form a reflector. In this regard, electromagnetic fields that are multiply reflected from the layers of the multi-layer dielectric stack **1612** may constructively interfere such that the multi-layer dielectric stack **1612** acts as a reflector. In some embodiments, the multi-layer dielectric stack **1612** may provide a strongly reflected photonic signal without ohmic loss that may otherwise occur using a metallic reflector **1610**.
[0158] As shown in FIGS. **16**G and **16**H, each of the photonic interconnect dies **502** may include a first photonic coupler **112***a* and a second photonic coupler **112***b* having a curved reflector surface **806**. As described above with reference to FIG. **8**C, a curved reflector surface **806** may be generated by performing an etching process using multi-tone masks. As shown in FIG. **16**H, the photonic interconnect die may further include a transition edge coupler **1616** structure. As shown, the transition edge coupler **1616** may include additional core portions (**524***a*, **524***b*) formed adjacent to the core portion **524**. The additional core portions (**524***a*, **524***b*) may be formed using processing operations similar to those described above with reference to FIGS. **14**A to **15**C. The transition edge coupler **1616** may act to expand a lateral width of an electromagnetic field associated with a photonic signal propagating within the core portion **524** of the dielectric waveguide **104**. Similar edge couplers **306***b* (see FIG. **3**D) are described with reference to FIGS. **17**A and **17**B, below.
[0159] FIG. **17**A is a top view of a photonic interconnect die **502** including edge couplers **306***b* (also see FIG. **3**D), and FIG. **17**B is a vertical cross-sectional view of the photonic interconnect die **502** of FIG. **17**A, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **17**B is indicated by the cross-section B-B′ in FIG. **17**A. As shown, the photonic interconnect die **502** may include a plurality of dielectric waveguides **104** that include a core portion **524** surrounded by a cladding portion **522**. As shown, each core portion **524** may have tapered ends **1702**. Each of the tapered ends **1702** of the core portions **524** may be enclosed withing a further region of a dielectric material **1704** to thereby form a spot size converter (SSC). As shown, a distribution of electromagnetic field lines **1706** may be expanded by the edge couplers **306***b*. In this regard, fields within the core portions may have a smaller lateral distribution than field lines that may be coupled into an out of the photonic interconnect die. As such, the photonic interconnect die **502**, including edge couplers **306***b*, may efficiently coupler photonic signals from a dielectric waveguide **104** to an external optical fiber (not shown).
[0160] As shown in FIG. **17**B, a first dielectric window **1604***a* and a second dielectric window **1604***b* may be formed at respective surfaces (**1608***b*, **1608***c*) of the first photonic coupler **112***a* and the second photonic coupler **112***b*. In this regard, the first dielectric window **1604***a* may be located on a second surface **1608***b* of the photonic interconnect die **502** that is perpendicular to the first surface **1608***a* of the photonic interconnect die **502**. Similarly, the second dielectric window **1604***b* may be located on a third surface **1608***c* of the photonic interconnect die **502** that is also perpendicular to the first surface **1608***a* of the photonic interconnect die **502**. In this example embodiment, the third surface **1608***c* of the photonic interconnect die **502** is parallel and opposite to the second surface **1608***b*. As such, the first photonic coupler **112***a* and the second photonic coupler **112***b* each couple photonic signals **310** into and out of the photonic interconnect die **502** through the first dielectric window **1604***a* and the second dielectric window **1604***b*, respectively.
[0161] FIG. **18**A is a top view of an intermediate structure that may be used to form a photonic

interconnect die **502** include grating couplers **306***a* (also see FIG. **3**C), FIG. **18**B is a vertical cross-sectional view of the intermediate structure of FIG. **18**A, according to various embodiments, and FIG. **18**C is a vertical cross-sectional view of a photonic interconnect die including grating couplers, according to various embodiments. The vertical plane defining the cross-sectional view of FIG. **18**B is indicated by the cross-section B-B′ in FIG. **18**A. As shown, each of a plurality of dielectric waveguides **104** may include a respective core portion **524** surrounded by a cladding portion. The dielectric waveguides of FIGS. **18**A to **18**C may be formed by processed similar to those described above with reference to FIGS. **14**A to **15**C.

[0162] As shown in FIG. **18**A, each core portion **524** may include grating couplers **306***a* formed at respective ends of the core portions **524**. In this regard, each end of a core portion **524** may include a flared portion that includes a plurality of grooves **308**. The grooves may be formed by performing an etching operation in top surfaces of the core portions **524**, as shown in FIG. **18**B. The photonic interconnect die **502** of FIG. **18**C may be formed from the intermediate structure **1800** of FIGS. **18**A and **18**B buy forming an additional cladding material layer such that the cladding portion **522** surrounds the core portion **524**. As shown in FIG. **18**C, dielectric windows (**1604***a*, **1604***b*) may then be formed by depositing a further dielectric material **1604**L (see FIG. **16**C) over the intermediate structure **1800**. Excess portions of the dielectric material layer **1604**L may then be removed by performing a planarization process to thereby generate the dielectric windows (**1604***a*, **1604***b*), as described with reference to FIGS. **16**C and **16**D, above.

[0163] As shown, photonic signals **310** may be coupled into and out from the photonic interconnect die **502** by the photonic couplers (**112***a*, **112***b*). Thus, as with the embodiments of FIGS. **16**D to **16**H, the first dielectric window **1604***a* may be located at a first position (e.g., the left side) on the first surface **1608***a* of the photonic interconnect die **502** and the second dielectric window **1604***b* may be located at a second position (e.g., on the right side) on the first surface **1608***a* of the photonic interconnect die **502**.

[0164] FIGS. **19**A to **19**D are vertical cross-sectional views of intermediate structures (**1900***a* to **1900***d*) that may be used to form a photonic interconnect die, according to various embodiments. The intermediate structure **1900***a* may be formed by forming a cladding material **522**L layer over a substrate (not shown). The cladding material layer **522**L may be chosen to be a photosensitive polymer material. A laser-writing operation may then be performed on the cladding material layer **522**L to thereby generate a core portion **524**. In this regard, as shown in FIG. **19**B, a laser **1902** may generate a beam of laser radiation **1904** that may be directed to the cladding material layer **522**L at a certain depth **1906** within the cladding material layer. The laser radiation **1904** may be chosen to have a certain beam width **1908**.

[0165] The interaction of the laser radiation **1904** with the cladding material layer **522**L may act to alter the micro-structure of the polymer material of the cladding material layer **522**L in a localized region in response to absorption of laser radiation during the laser-writing operation. The altered microstructure may be characterized by an increased refractive index relative to the cladding material layer **522**L. As such, the irradiated localized region may then serve as the core portion **524** while a surrounding un-irradiated portion of the cladding material layer **522**L may serve as the cladding portion **522**, as shown in FIG. **19**C. The use of a laser-writing operation may allow considerable flexibility regarding a spatial layout of core portions **524**. For example, as shown in FIG. **19**D, a plurality of dielectric waveguides **104** may be generated having a non-uniform spatial distribution. For example, the plurality of dielectric waveguides **104** may include a fan-out configuration in which a linear spacing of core portions **524** may laterally increase (e.g., in the y-direction) as a function of a longitudinal direction (e.g., the x-direction).

[0166] FIG. **20** is a flowchart illustration operations of a method **2000** of forming a photonic interconnect die **502**, according to various embodiments. In operation **2002**, the method **2000** may include forming a waveguide cladding portion **522** on a substrate **520**. In operation **2004**, the method **2000** may include forming a waveguide core portion **524** within the waveguide cladding portion **522**. In operation **2006**, the method **2000** may include forming a first photonic coupler **112***a* at a first end of the waveguide core portion **524**. In operation **2008**, the method **2008** may include forming a second photonic coupler **112***b* at a second end of the waveguide core portion **524**.

[0167] In forming the cladding portion (**522**, **304**) in operation **2002**, and forming the waveguide core portion (**524**, **302***b*) in operation **2004**, the method **2000** may further include forming a silicon-on-insulator substrate **302** including a silicon substrate **302***a*, a first silicon dioxide layer **304** formed over the silicon substrate **302***a*, and a silicon layer **302***b* formed over the first silicon dioxide layer **304**. The method **2000** may further include patterning and etching the silicon layer **302***b* to form a silicon waveguide core portion (**524**, **302***b*). The method **2000** may further include forming a second silicon dioxide layer **304** over the waveguide core portion (**522**, **304**) such that the waveguide core portion **524** is surrounded by silicon dioxide **304** so that the waveguide cladding portion **522** may include the first silicon dioxide layer **304** and the second silicon dioxide layer **304**.

[0168] In forming the cladding portion (**522**, **304**) in operation **2002**, and forming the waveguide core portion (**524**, **302***b*) in operation **2004**, the method **2000** may further include forming a first layer of a first polymer material **522** over the substrate **520**; forming a second layer of a second polymer material **524**L over the substrate **520**; pattering the second layer of the second polymer material **524**L to form the waveguide core portion **524**; and forming a third layer of the first polymer material **522** over the waveguide core portion **524**, such that the first layer and the third layer of the first polymer material **522** form the waveguide cladding portion **522**. In forming the cladding portion (**522**, **304**) in operation **2002**, and forming the waveguide core portion (**524**, **302***b*) in operation **2004**, the method **2000** may further include forming a radiation-curable polymer material **522**L over the substrate **520**; and irradiating a region of the radiation-curable polymer material **522**L with laser radiation **1904** in a laser-writing operation to thereby form the waveguide core portion **524** of a waveguide, such that an un-radiated portion of the radiation-curable polymer material **522**L may serve as the waveguide cladding portion **522**.

[0169] Referring to all drawings and according to various embodiments of the present disclosure, a photonic interconnect die **502** is provided. The photonic interconnect die **502** may include a substrate **520**, a dielectric waveguide **104**, including a core portion **524** and a cladding portion **522**, formed on the substrate **520**, a first photonic coupler **112***a* formed at a first end of the dielectric waveguide **104**, and a second photonic coupler **112***b* formed at a second end of the dielectric waveguide **104**. The dielectric waveguide **104** may include a planar geometry within the cladding portion **522** such that a surface of the dielectric waveguide **104** is parallel to a first surface **1608***a* of the photonic interconnect die **502**.

[0170] The first photonic coupler **112***a* and the second photonic coupler **112***b* may each couple photonic signals **310** into and out of the photonic interconnect die **502** such that a photonic signal pathway (**514***a*, **524**, **514***b*) connects the first photonic coupler **112***a*, the dielectric waveguide **104**, and the second photonic coupler **112***b*. The photonic interconnect die **502** may further include a first dielectric window **1604***a* located at a first position on the first surface **1608***a* of the photonic interconnect die **502** and a second dielectric window **1604***b* located at a second position on the first surface **1608***a* of the photonic interconnect die **502**. The first photonic coupler **112***a* and the second photonic coupler **112***b* may couple photonic signals **310** into and out of the photonic interconnect die **502** through the first dielectric window **1604***a* and the second dielectric window **1604***b*, respectively.

[0171] In some embodiments, the first photonic coupler **112***a* may include a first grating coupler **306***a* that couples first input photonic signals **310** into the dielectric waveguide **104** that are received from the first dielectric window **1604***a* and transmits first output photonic signals **310** through the first dielectric window **1604***a* that are received from the dielectric waveguide **104**. Similarly, the second photonic coupler **112***b* may include a second grating coupler **306***a* that couples second input photonic signals **310** into the dielectric waveguide **104** that are received from the second dielectric window **1604***b* and transmits second output photonic signals **310** through the second dielectric window **1604***b* that are received from the dielectric waveguide **104**.

[0172] In further embodiments, the first photonic coupler **112***a* may include a first angled reflector **306***c* that couples first input photonic signals **310** into the dielectric waveguide **104** that are received from the first dielectric window **1604***a* and transmits first output photonic signals **310** through the first dielectric window **1604***a* that are received from the dielectric waveguide **104**. Similarly, the second photonic coupler **112***b* may include a second angled reflector **306***c* that couples second input photonic signals **310** into the dielectric waveguide **104** that are received from the second dielectric window **1604***b* and transmits second output photonic signals **310** through the second dielectric window **1604***b* that are received from the dielectric waveguide **104**.

[0173] In other embodiments, the first dielectric window **1604***a* may be located on a second surface **1608***b* of the photonic interconnect die **502** that is perpendicular to the first surface **1608***a* of the photonic interconnect die **502** and the second dielectric window **1604***b* may be located on a third surface **1608***c* of the photonic interconnect die **502** that is perpendicular to the first surface **1608***a* of the photonic interconnect die **502** such that the third surface **1608***c* of the photonic interconnect die **502** is parallel and opposite to the second surface **1608***b*. In such embodiments, the first photonic coupler **112***a* and the second photonic coupler **112***b* may each couple the photonic signals **310** into and out of the photonic interconnect die **502** through the first dielectric window **1604***a* and the second dielectric window **1604***b*, respectively.

[0174] According to certain embodiments, the first photonic coupler **112***a* may include a first edge coupler **306***b* that couples first input photonic signals **310** into the dielectric waveguide **104** that are received from the first dielectric window **1604***a* and transmits first output photonic signals **310** through the first dielectric window **1604***a* that are received from the dielectric waveguide **104**. Similarly, the second photonic coupler **112***b* may include a second edge coupler **306***b* that couples second input photonic signals **310** into the dielectric waveguide **104** that are received from the second dielectric window **1604***b* and transmits second output photonic signals **310** through the second dielectric window **1604***b* that are received from the dielectric waveguide **104**. In some embodiments, each of the first edge coupler **306***b* and the second edge coupler **306***b* may include a tapered end **1702** of the dielectric waveguide **104** in contact with an enclosing dielectric material **1704**.

[0175] According to certain embodiments, each of the first edge coupler **306***b* and the second edge coupler **306***b* may include a transition edge coupler **1616**. The core portion **524** may include a first material having a first index of refraction and the cladding portion **522** may include a second material having a second index of refraction that is less than the first index of refraction. In some embodiments, the core portion **524** may include silicon and the cladding portion **522** may include silicon dioxide. In other embodiments, the core portion **524** may include a first polymer material and the cladding portion **522** may include a second polymer material.

[0176] Referring to all drawings and according to various embodiments of the present disclosure, an electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) is provided. The electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) may include a first photonic component (**200***a*, **402***a*) including first photonic signal pathways (**310**, **514***a*), a second photonic component (**200***b*, **402***b*) including second photonic signal pathways (**310***b*, **514***b*), and a photonic interconnect die **502** including a plurality of dielectric waveguides **104**. The photonic interconnect die **502** may be coupled to the first photonic component (**200***a*, **402***a*) and the second photonic component (**200***b*, **402***b*) such that the first photonic signal pathways (**310**, **514***a*) are photonically coupled to the second photonic signal pathways (**310***b*, **514***b*) by the plurality of dielectric waveguides **104**.

[0177] The electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) may further include an interposer **408**, such that the first photonic component (**200***a*, **402***a*) and the second photonic component (**200***b*, **402***b*) are formed as separate dies and are attached to, and electrically coupled to, the interposer **408**. A first portion **526***a* of the photonic interconnect die **502** may be mechanically and photonically coupled to the first photonic component (**200***a*, **402***a*) and a second portion **526***b* of the photonic interconnect die **502** may be mechanically and photonically coupled to the second photonic component (**200***b*, **402***b*) so that the photonic interconnect die **502** is configured as an intra-package photonic coupler (**502***a*, **502***c*, see FIG. **8**B).

[0178] In further embodiments, the electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) may include a first interposer **408***a*, and a second interposer **408***b*, such that the first photonic component (**200***a*, **402***a*) is attached to, and electrically coupled to, the first interposer **408***a*, and the second photonic component (**200***b*, **402***b*) is attached to, and electrically coupled to, the second interposer **408***b*. In this regard, a first portion **526***a* of the photonic interconnect die **502** may be mechanically and photonically coupled to the first photonic component (**200***a*, **402***a*) and a second portion **526***b* of the photonic interconnect die **502** may be mechanically and photonically coupled to the second photonic component (**200***b*, **402***b*) so that the photonic interconnect die **502** is configured as an inter-package photonic coupler **502***b* (see FIG. **8**B).

[0179] According to various embodiments, a photonic interconnect die **502** may further include a substrate **520**, a cladding portion **522** formed on the substrate **520**, a plurality of dielectric waveguide core portions **524** formed within the cladding portion a plurality of first photonic couplers **112***a*, and a plurality of second photonic couplers **112***b*. According to various embodiments, respective ones of the plurality of first photonic couplers **112***a* and the plurality of second photonic couplers **112***b* are coupled to first and second ends of respective ones of the plurality of dielectric waveguide core portions **524**. Further, the plurality of first photonic couplers **112***a* and the plurality of second photonic couplers **112***b* may be configured to guide photonic signals **310** into and out of the photonic interconnect die **502** such that a respective photonic signal pathway (**310***c*, **524**) connects each of the plurality of dielectric waveguides **104** with respective ones of the plurality of first photonic couplers **112***a* and the plurality of second photonic couplers **112***b*. In some embodiments, the plurality of dielectric waveguides **104** may be formed on a common planar substrate **520** and may include a fan-out configuration (see FIG. **19**D). Alternatively, the plurality of dielectric waveguides **104** may be arranged in a three-dimensional configuration (see FIG. **15**C) within the cladding portion **522**.

[0180] Disclosed embodiments provide photonic interconnect dies **502** that allow photonic signals **310** to be propagated between a first photonic component (**200***a*, **402***a*) and a second photonic component (**200***b*, **402***b*) in an electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**). The incorporation of optical/photonic signaling functionality into an electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) may provide reduced signal delay and ohmic loss. In this regard, in certain embodiments, to avoid long electrical signal pathways and associated delay and ohmic loss, it may be advantageous to convey signals in the form of photonic signals **310** along a portion of the signal pathway.

[0181] This may be done by converting electrical signals into photonic signals **310** at a first point along the signal pathway, propagating the photonic signals for a certain distance, and then converting the photonic signals back into electrical signals at a second point along the signal pathway. In still further embodiments, the functionality to convert electrical signals into photonic signals **310** and vice versa may be advantageous in photonic (quantum or classical) computing operations. The use of photonic interconnect dies **502** allows various electronic and photonic components (**200**, **402**) to be fabricated separately as stand-alone dies. Such dies may then be assembled into an electronic/photonic package (**400**, **500**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1300**) and photonic components may be coupled by photonic interconnect dies **502**.

[0182] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure

Claims

1. A photonic interconnect die, comprising: a substrate; a dielectric waveguide, comprising a core portion and a cladding portion, formed on the substrate; a first photonic coupler formed at a first end of the dielectric waveguide; and a second photonic coupler formed at a second end of the dielectric waveguide, wherein the dielectric waveguide comprises a planar geometry within the cladding portion such that a surface of the dielectric waveguide is parallel to a first surface of the photonic interconnect die, and wherein the first photonic coupler and the second photonic coupler each couple photonic signals into and out of the photonic interconnect die such that a photonic signal pathway connects the first photonic coupler, the dielectric waveguide, and the second photonic coupler.
2. The photonic interconnect die of claim 1, further comprising: a first dielectric window located at a first position on the first surface of the photonic interconnect die; and a second dielectric window located at a second position on the first surface of the photonic interconnect die, wherein the first photonic coupler and the second photonic coupler each couple photonic signals into and out of the photonic interconnect die through the first dielectric window and the second dielectric window, respectively.
3. The photonic interconnect die of claim 2, wherein: the first photonic coupler comprises a first grating coupler that couples first input photonic signals into the dielectric waveguide that are received from the first dielectric window and transmits first output photonic signals through the first dielectric window that are received from the dielectric waveguide; and the second photonic coupler comprises a second grating coupler that couples second input photonic signals into the dielectric waveguide that are received from the second dielectric window and transmits second output photonic signals through the second dielectric window that are received from the dielectric waveguide.
4. The photonic interconnect die of claim 2, wherein: the first photonic coupler comprises a first angled reflector that couples first input photonic signals into the dielectric waveguide that are received from the first dielectric window and transmits first output photonic signals through the first dielectric window that are received from the dielectric waveguide; and the second photonic coupler comprises a second angled reflector that couples second input photonic signals into the dielectric waveguide that are received from the second dielectric window and transmits second output photonic signals through the second dielectric window that are received from the dielectric waveguide.
5. The photonic interconnect die of claim 1, further comprising: a first dielectric window located on a second surface of the photonic interconnect die that is perpendicular to the first surface of the photonic interconnect die; and a second dielectric window located on a third surface of the photonic interconnect die that is perpendicular to the first surface of the photonic interconnect die such that the third surface of the photonic interconnect die is parallel and opposite to the second surface, wherein the first photonic coupler and the second photonic coupler each couple the photonic signals into and out of the photonic interconnect die through the first dielectric window and the second dielectric window, respectively.
6. The photonic interconnect die of claim 5, wherein: the first photonic coupler comprises a first edge coupler that couples first input photonic signals into the dielectric waveguide that are received from the first dielectric window and transmits first output photonic signals through the first dielectric window that are received from the dielectric waveguide; and the second photonic coupler

comprises a second edge coupler that couples second input photonic signals into the dielectric waveguide that are received from the second dielectric window and transmits second output photonic signals through the second dielectric window that are received from the dielectric waveguide.

7. The photonic interconnect die of claim 6, wherein each of the first edge coupler and the second edge coupler comprises a tapered end of the dielectric waveguide in contact with an enclosing dielectric material.

8. The photonic interconnect die of claim 6, wherein each of the first edge coupler and the second edge coupler comprises a transition edge coupler.

9. The photonic interconnect die of claim 1, wherein the core portion comprises a first material having a first index of refraction and the cladding portion comprises a second material having a second index of refraction that is less than the first index of refraction.

10. The photonic interconnect die of claim 1, wherein the core portion comprises silicon and the cladding portion comprises silicon dioxide.

11. The photonic interconnect die of claim 1, wherein the core portion comprises a first polymer material and the cladding portion comprises a second polymer material.

12. An electronic/photonic package, comprising: a first photonic component comprising first photonic signal pathways; a second photonic component comprising second photonic signal pathways; and a photonic interconnect die comprising a plurality of dielectric waveguides, wherein the photonic interconnect die is coupled to the first photonic component and the second photonic component such that the first photonic signal pathways are photonically coupled to the second photonic signal pathways by the plurality of dielectric waveguides.

13. The electronic/photonic package of claim 12, further comprising: an interposer, wherein the first photonic component and the second photonic component are formed as separate dies and are attached to, and electrically coupled to, the interposer, and wherein a first portion of the photonic interconnect die is mechanically and photonically coupled to the first photonic component and a second portion of the photonic interconnect die is mechanically and photonically coupled to the second photonic component so that the photonic interconnect die is configured as an intra-package photonic coupler.

14. The electronic/photonic package of claim 12, further comprising: a first interposer; and a second interposer, wherein: the first photonic component is attached to, and electrically coupled to, the first interposer; and the second photonic component is attached to, and electrically coupled to, the second interposer, and wherein a first portion of the photonic interconnect die is mechanically and photonically coupled to the first photonic component and a second portion of the photonic interconnect die is mechanically and photonically coupled to the second photonic component so that the photonic interconnect die is configured as an inter-package photonic coupler.

15. The electronic/photonic package of claim 12, wherein the photonic interconnect die further comprises: a substrate; a cladding portion formed on the substrate; a plurality of dielectric waveguide core portions formed within the cladding portion; a plurality of first photonic couplers; and a plurality of second photonic couplers, wherein respective ones of the plurality of first photonic couplers and the plurality of second photonic couplers are coupled to first and second ends of respective ones of the plurality of dielectric waveguide core portions, and wherein the plurality of first photonic couplers and the plurality of second photonic couplers guide photonic signals into and out of the photonic interconnect die such that a respective photonic signal pathway connects each of the plurality of dielectric waveguides with respective ones of the plurality of first photonic couplers and the plurality of second photonic couplers.

16. The electronic/photonic package of claim 15, wherein: the plurality of dielectric waveguides are formed on a common planar substrate and comprise a fan-out configuration, or the plurality of dielectric waveguides is arranged in a three-dimensional configuration within the cladding portion.

17. A method of forming a photonic interconnect die, comprising: forming a waveguide cladding portion on a substrate; forming a waveguide core portion within the waveguide cladding portion; forming a first photonic coupler at a first end of the waveguide core portion; and forming a second photonic coupler at a second end of the waveguide core portion.

18. The method of claim 17, wherein forming the waveguide cladding portion on the substrate and forming the waveguide core portion within the waveguide cladding portion further comprises: forming a silicon-on-insulator substrate comprising a silicon substrate, a first silicon dioxide layer formed over the silicon substrate and a silicon layer formed over the first silicon dioxide layer; patterning and etching the silicon layer to form a silicon waveguide core portion; and forming a second silicon dioxide layer over the waveguide core portion such that the waveguide core portion is surrounded by silicon dioxide so that the waveguide cladding portion comprises the first silicon dioxide layer and the second silicon dioxide layer.

19. The method of claim 17, wherein forming the waveguide cladding portion on the substrate and forming the waveguide core portion within the waveguide cladding portion further comprises: forming a first layer of a first polymer material over the substrate; forming a second layer of a second polymer material over the substrate; pattering the second layer of the second polymer material to form the waveguide core portion; and forming a third layer of the first polymer material over the waveguide core portion, wherein the first layer and the third layer of the first polymer material comprise the waveguide cladding portion.

20. The method of claim 17, wherein forming the waveguide cladding portion on the substrate and forming the waveguide core portion within the waveguide cladding portion further comprises: forming a radiation-curable polymer material over the substrate; and irradiating a region of the radiation-curable polymer material with laser radiation in a laser-writing operation to thereby form the waveguide core portion of a waveguide, wherein an un-radiated portion of the radiation-curable polymer material comprises the waveguide cladding portion.